Finite Sample Bounds for Composite Hypothesis Testing

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Abstract

We investigate composite binary hypothesis testing in the finite sample regime under asymmetric error constraints. Using Rényi divergences, we derive explicit achievability and converse bounds for the optimal Type II error. When the Type I error is constrained to decay exponentially with sample size, the bounds identify a phase transition and yield a strong converse above it. In the composite problem, the phase transition threshold is given by the joint KL projection over the alternative and null classes. Achievability is obtained through a joint Rényi projection whose log likelihood ratio defines a single test with uniform error control over both hypothesis classes, without requiring the projected pair to be least favourable. For compact convex classes with full support on a finite alphabet, we determine the exact error exponents on both sides of the transition and show that the achievable exponent is attained at a unique Rényi order. The same framework recovers the fixed Type I composite Chernoff–Stein exponent and yields a polynomial refinement of the finite sample achievability result. We further identify conditions under which the projected pair is least favourable at finite sample size.

Index Terms

Composite Hypothesis Testing, Minimax Testing, Finite Sample, Rényi Divergence, Rényi projection, Type II Error Exponent, Strong Converse, Least Favourable Distributions, Robust Detection

I. Introduction

Binary hypothesis testing asks whether observed data were generated under a null hypothesis or an alternative hypothesis. In the Neyman Pearson (NP) formulation, the probability of rejecting a true null hypothesis is constrained, while the probability of accepting a false null hypothesis is minimised [1], [2]. These are the Type I and Type II errors, respectively.

When the null and alternative hypotheses each specify a single probability law, the problem is a simple hypothesis test, and the NP lemma identifies an optimal likelihood ratio test [1]. Under a fixed Type I error constraint, the asymptotic behaviour of the optimal Type II error is characterised by the Chernoff-Stein lemma, which gives the Kullback Leibler (KL) divergence as the optimal Type II error exponent [3]. If instead the Type I error is required to decay exponentially as a function of sample size, the asymptotic behaviour depends on an imposed rate r.

Here the problem exhibits a phase transition. The Type II error decays exponentially to zero when r is below the phase transition threshold, while above it the Type II error converges exponentially to one. The Type II error decays exponentially to zero below a critical Type I error rate, while above this rate it converges exponentially to one. The threshold separating these two regimes is the KL divergence from the alternative hypothesis to the null hypothesis [4], [5]. Bruno et al. [6] recently derived finite sample achievability and converse bounds for the simple binary hypothesis testing problem using Rényi divergences and recovered this classical threshold rate. They thus provided an explicit achievable exponential rate for the Type II error below it.

In this work, we study the problem when both hypotheses are composite. The null and alternative hypotheses each comprise a class of probability laws, and the test must be chosen without knowing which member of either class generated the data. Consequently, the Type I error constraint must hold for every distribution in the null class, while the Type II error is assessed over the alternative class. Unlike in simple hypothesis testing, the hypotheses no longer determine a single likelihood ratio because they do not specify a single null and alternative distribution. One standard approach is the generalised likelihood ratio test, which compares the maximised likelihoods under the two classes [7].Although widely used, the GLR test is not generally optimal for composite hypothesis testing [7]. Another classical approach uses least favourable distributions to reduce a composite testing problem to a simple binary test [8].

Robust detection provides another classical approach. Here one seeks least favourable distributions that reduce the composite problem to a simple binary test [8]. Under structural conditions such as stochastic ordering, Choquet capacities, or Blackwell dominance, a pair can be identified that simultaneously attains the extremal Type I and Type II errors over the two classes [8]–[10]. The resulting likelihood ratio test is then NP optimal for the composite problem at every sample size. Such a pair is not guaranteed to exist for general composite classes, and classical existence results require additional structural conditions. Finite sample least favourability is also stronger than determining the optimal asymptotic error exponent. A pair may determine this exponent without attaining the extremal error probabilities at finite sample size. Gül [11] illustrates this distinction for convex uncertainty classes.

Other studies characterise asymptotic error exponents for structured composite problems. Tomamichel $\&$ Hayashi [12] consider asymmetric testing with a fixed i.i.d. null distribution and a structured composite alternative. Their framework uses a sequence of universal distributions on the sample space that dominates the admissible alternatives up to a factor that grows polynomially with sample size. This construction permits a single test to control the composite alternative and yields the critical exponential Type II error rate separating the error exponent and strong converse regimes, together with the corresponding exact exponents.

For product and Markov alternative classes, these exponents admit operational interpretations in terms of variants of Sibson-α mutual information and Rényi conditional mutual information. They also derive a Gaussian second order expansion around this critical rate on the $\sqrt{n}$ scale. This concerns local deviations around the critical rate and is distinct from the fixed large deviation refinement considered here, where polynomial factors modify the dominant exponential behaviour.

Shayevitz [13] gives a different operational interpretation of Rényi divergence through a two sensor composite testing problem, showing under an extremal noise condition that the optimal miss detection exponent is determined by a Rényi divergence between the corresponding distribution families. Huang and Moulin [14] study a simple null against a finite collection of product alternatives and derive strong large deviation asymptotics for the achievable error region, determining it up to constant order. Moulin and Johnstone [15] consider a known null against a regular finite dimensional exponential family alternative and obtain sharp asymptotics for Rao and generalised likelihood ratio tests, including dimension dependent higher order terms. In the quantum setting, Berta, Brandão, and Hirche [16] extend Stein's lemma to convex combinations of i.i.d. quantum states and show that the optimal Type II error exponent is generally given by a regularised quantum relative entropy, which reduces to a nonregularised expression in particular cases. Mosonyi, Szilágyi, and Weiner [17] give a broader analysis of composite error exponents, identifying conditions under which the optimal composite exponent coincides with the worst pairwise exponent and examples in which this reduction fails.

These results address important structured instances of composite testing, but none provide finite sample achievability and converse bounds for the classical setting considered here, where both hypotheses are composite and a single test is subject to a uniform Type I error constraint over the null class while the Type II error is evaluated over the alternative class. Our work provides explicit bounds for this problem.

A. Contributions

Let $\mathcal{C}_0$ and $\mathcal{C}_1$ denote the null and alternative classes, and write $\beta_n^\star(\varepsilon)$ for the optimal Type II error under the uniform Type I constraint ε.

Our main contributions are as follows.

1) We derive a finite sample converse for the composite problem. For every admissible test and every $P \in \mathcal{C}_0$, $Q \in \mathcal{C}_1$, pairwise reduction gives a lower bound on $\beta_n^\star(\varepsilon; \mathcal{C}_0, \mathcal{C}_1)$ in terms of the corresponding simple binary problem (P, Q). This yields an explicit composite Rényi converse without assuming the existence of a least favourable pair. See Section III-A.
2) We derive a finite sample achievability bound without assuming that the composite problem admits a least favourable pair. For $\lambda \in (0,1)$, a joint Rényi projection $(Q_\lambda^\star, P_\lambda^\star)$ attaining $\inf\limits_{Q\in\mathcal{C}_1} \inf\limits_{P\in\mathcal{C}_0} D_\lambda(Q\|P)$ defines a single test through the log-likelihood ratio $\log(Q_\lambda^\star/P_\lambda^\star)$. We show that this test controls the Type I error uniformly over $\mathcal{C}_0$ and the Type II error uniformly over $\mathcal{C}_1$. Again, this does not assume least favourability of the distribution and the likelihood-ratio test matches the least favourable one only in certain cases. See Section III-B.
3) Under the exponentially decaying Type I constraint $\varepsilon = e^{-nr}$, we extend the classical phase transition for simple binary testing to the composite setting. The threshold is given by the KL projection of the alternative class onto the null class, $r_c = \inf_{Q\in\mathcal{C}_1} \inf_{P\in\mathcal{C}_0} D(Q\|P)$. For $r < r_c$, $\beta_n^\star(e^{-nr})$ decays exponentially to zero, whereas for $r > r_c$ it converges to one. For compact convex classes with full support on a finite alphabet, we determine the exact exponents on both sides of this transition. In the achievable regime, the Type II error exponent is the smallest exponent among all simple pairs. In the converse regime, the exponent governing convergence to one is the largest strong converse exponent among all simple pairs. See Sections III-C, IV-A and IV-B.

4) For compact convex classes with full support on a finite alphabet, we show that our finite sample Rényi achievability bound recovers the known composite Chernoff–Stein exponent under a fixed Type I constraint. Specifically, the exponent $\inf_{\substack{P\in\mathcal{C}_0\\Q\in\mathcal{C}_1}} D(P\|Q)$ emerges from the order zero limit of the Rényi achievability expression, while the order one limit gives the phase transition threshold $\inf_{\substack{Q\in\mathcal{C}_1\\P\in\mathcal{C}_0}} D(Q\|P)$. Thus the Rényi formulation connects the fixed and exponentially decaying Type I regimes through its order zero and order one limits. Moreover, the achievable exponent for $\varepsilon = e^{-nr}$ converges to the fixed Type I exponent as $r \downarrow 0$, giving a continuous connection between the two regimes at zero rate. See Section IV-C.
5) We analyse the achievability result beyond the exponential rate. We show that the optimal Type II error has a polynomial prefactor $n^{-1/(2\lambda_r^\star)}$ multiplying its dominant exponential decay. We also optimise the likelihood ratio test induced by the Rényi projection under the exact Type I constraint, prove uniqueness of the maximising Rényi order $\lambda_r^\star$, and identify the corresponding logarithmic threshold correction $-(2\lambda_r^\star)^{-1}\log n + O(1)$. See Sections IV-A, V-A and V-B.
6) We ask whether the pair $(Q_\lambda^\star, P_\lambda^\star)$, selected by the Rényi projection, can also be least favourable at finite sample size. We give sufficient stochastic ordering conditions on $\log(Q_\lambda^\star/P_\lambda^\star)$ under which the composite problem reduces exactly to the corresponding simple binary problem. We then show that these conditions hold for separated one parameter natural exponential families, where the Rényi projection gives the least favourable pair. See Section VI.

II. Problem formulation and definitions

A. Problem formulation

Let $\mathcal{C}_0$ and $\mathcal{C}_1$ be nonempty classes of probability laws on a common measurable space $(\mathcal{X}, \mathcal{F})$. For a sample size $n \geq 1$, let $X^n = (X_1, \dots, X_n)$. The composite binary hypothesis testing problem is

$$H_{\mathcal{C}_0} : X^n \sim P^{\otimes n} \quad \text{for some } P \in \mathcal{C}_0, \quad \text{vs.} \quad H_{\mathcal{C}_1} \quad : X^n \sim Q^{\otimes n} \quad \text{for some } Q \in \mathcal{C}_1.$$

Let $\varphi_n : \mathcal{X}^n \to [0,1]$ be an $\mathcal{F}^{\otimes n}$ measurable randomised test at sample size n. Randomisation may be necessary to attain the Type I constraint exactly at finite sample size, since a deterministic decision rule may not achieve an arbitrary prescribed value of ε. Moreover, randomisation at the threshold permits this exact calibration. For an observation x^n, $\varphi_n(x^n)$ is the probability of deciding $H_{\mathcal{C}_1}$. Its Type I error over the null class is

$$\alpha_n(\varphi_n; \mathcal{C}_0) := \sup_{P\in\mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}\left[\varphi_n(X^n)\right].$$

For $\varepsilon \in (0,1)$, define

$$\beta_n^\star(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) := \inf_{\varphi_n :\, \alpha_n(\varphi_n;\mathcal{C}_0)\leq\varepsilon} \sup_{Q\in\mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}\left[1 - \varphi_n(X^n)\right]. \tag{1}$$

This is the optimal Type II error at sample size n under the uniform Type I error constraint ε. The infimum is taken over all randomised tests satisfying this constraint.

B. Definitions

Definition 1: Let P and Q be probability laws on $(\mathcal{X}, \mathcal{F})$. For $\lambda > 0$, $\lambda \neq 1$, let μ be any measure with respect to which both P and Q admit densities. The order-λ Hellinger integral [18] is

$$H_\lambda(Q, P) := \int_{\mathcal{X}} \left(\frac{dQ}{d\mu}\right)^{\lambda} \left(\frac{dP}{d\mu}\right)^{1-\lambda} d\mu, \tag{2}$$

and the Rényi divergence [18] of order λ is

$$D_\lambda(Q\|P) := \frac{1}{\lambda - 1} \log H_\lambda(Q, P). \tag{3}$$

These quantities do not depend on the choice of μ. Standard extended real conventions apply, and $D(Q\|P)$ denotes the KL divergence.

Definition 2: For $\lambda > 0$, $\lambda \neq 1$, define

$$D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) := \inf_{Q\in\mathcal{C}_1} \inf_{P\in\mathcal{C}_0} D_\lambda(Q\|P), \quad D(\mathcal{C}_1\|\mathcal{C}_0) := \inf_{Q\in\mathcal{C}_1} \inf_{P\in\mathcal{C}_0} D(Q\|P).$$

Definition 3: For $\lambda \in (0, 1)$, a pair $(Q^\star_\lambda, P^\star_\lambda) \in \mathcal{C}_1 \times \mathcal{C}_0$ is called a joint Rényi projection if

$$D_\lambda(Q^\star_\lambda\|P^\star_\lambda) = \inf_{Q\in\mathcal{C}_1} \inf_{P\in\mathcal{C}_0} D_\lambda(Q\|P) = D_\lambda(\mathcal{C}_1\|\mathcal{C}_0).$$

We refer to $(Q^\star_\lambda, P^\star_\lambda)$ as the *projected pair*. Since $\lambda < 1$, this is equivalent to $(Q^\star_\lambda, P^\star_\lambda) \in \underset{(Q,P)\in\mathcal{C}_1\times\mathcal{C}_0}{\arg\max} H_\lambda(Q, P)$.

III. Bounds

We now derive finite sample bounds on the optimal Type II error (1) under a fixed Type I error constraint ε.

A. Converse bound

For the converse bound, any test admissible for the composite problem is also admissible for every simple binary problem obtained by fixing $P \in \mathcal{C}_0$ and $Q \in \mathcal{C}_1$. Applying the simple binary Rényi converse in [6, Thm. 1] to each such pair, then taking the infimum over the two classes and optimising over $\lambda > 1$, gives the following composite bound.

The following theorem gives a finite sample lower bound on $\beta^\star_n(\varepsilon; \mathcal{C}_0, \mathcal{C}_1)$ using a Rényi divergence of order $\lambda > 1$.

Theorem 1: For every $\varepsilon \in (0, 1)$ and $n \geq 1$,

$$\beta^\star_n(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) \geq 1 - \exp\left\{-\sup_{\lambda>1} \frac{\lambda-1}{\lambda}\left[\log\frac{1}{\varepsilon} - n \inf_{Q\in\mathcal{C}_1} \inf_{P\in\mathcal{C}_0} D_\lambda(Q\|P)\right]_+\right\}, \tag{4}$$

where $[a]_+ := \max\{a, 0\}$, with $[-\infty]_+ := 0$.

The proof of Theorem 1 is given in Appendix A-A.

B. Achievability bound

Constructing an achievability bound requires a single test that controls, uniformly, the Type I error over the null class and the Type II error over the alternative class. We first identify conditions under which a single statistic provides this uniform control.

Lemma 1: Fix $n \geq 1$, $\varepsilon \in (0,1)$, $\lambda \in (0,1)$, and $\xi_0, \xi_1 \geq 0$. Suppose that a measurable function $h : \mathcal{X} \to [-\infty, +\infty]$ satisfies

$$\sup_{P \in \mathcal{C}_0} \mathbb{E}_P \left[e^{\lambda h}\right] \leq e^{(\lambda-1)\xi_0}, \qquad \sup_{Q \in \mathcal{C}_1} \mathbb{E}_Q \left[e^{(\lambda-1)h}\right] \leq e^{(\lambda-1)\xi_1}.$$

For every

$$\tau \geq \frac{\log(1/\varepsilon) - n(1-\lambda)\xi_0}{\lambda}, \qquad \psi_{n,\tau}(x^n) = \mathbb{1}\left\{\sum_{i=1}^{n} h(x_i) \geq \tau\right\}, \tag{5}$$

satisfying $\alpha_n(\psi_{n,\tau}; \mathcal{C}_0) \leq \varepsilon$, and

$$\beta_n(\psi_{n,\tau}; \mathcal{C}_1) := \sup_{Q \in \mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}} \left[1 - \psi_{n,\tau}(X^n)\right] \leq \exp\left\{(1-\lambda)\tau + n(\lambda-1)\xi_1\right\}.$$

The proof of Lemma 1 is given in Appendix A-B.

The role of Lemma 1 is to isolate what is required for achievability in the composite setting. A single statistic is sufficient if its two exponential moments can be controlled uniformly over the two classes. For a fixed pair P and Q, taking $h = \log(Q/P)$ gives $\mathbb{E}_P[e^{\lambda h}] = \mathbb{E}_Q[e^{(\lambda-1)h}] = H_\lambda(Q, P)$. These moments are therefore determined by the Rényi divergence through (3). Setting the threshold to the smallest value allowed in (5) recovers the simple binary finite sample Rényi achievability bound in [6, Thm. 2].

The composite problem is therefore to find a single pair whose log-likelihood ratio provides this exponential control uniformly over the two classes. We select a pair that maximises the Hellinger integral over the classes. For $\lambda \in (0,1)$, this is equivalent to minimising the Rényi divergence, and hence to finding a joint Rényi projection $(Q_\lambda^\star, P_\lambda^\star)$. The convexity of the hypothesis classes, together with the concavity of the Hellinger integral, allows us to show that the log-likelihood ratio of this pair satisfies the required uniform bounds.

Throughout this section, weak compactness and weak semicontinuity in $L^1(\mu)$ are understood with respect to the weak topology $\sigma(L^1(\mu), L^\infty(\mu))$, under which $f_n \to f \iff \int f_n g \, d\mu \to \int f g \, d\mu, \ \forall g \in L^\infty(\mu)$.

Theorem 2: Fix $\lambda \in (0,1)$. Assume that all laws in $\mathcal{C}_0$ and $\mathcal{C}_1$ admit densities with respect to a common σ finite measure μ, and that the two classes are convex and weakly compact as defined above. Then a joint Rényi projection $(Q_\lambda^\star, P_\lambda^\star)$ exists. If one can be chosen such that $H_\lambda(Q_\lambda^\star, P_\lambda^\star) > 0$ and $R \ll P_\lambda^\star + Q_\lambda^\star$ for all $R \in \mathcal{C}_0 \cup \mathcal{C}_1$, then, for every $n \geq 1$ and $\varepsilon \in (0,1)$,

$$\beta_n^\star(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) \leq \min\left\{1 - \varepsilon, \exp\left[-\frac{1-\lambda}{\lambda}\left(n \inf_{Q \in \mathcal{C}_1} \inf_{P \in \mathcal{C}_0} D_\lambda(Q \| P) - \log\frac{1}{\varepsilon}\right)\right]\right\}. \tag{6}$$

Moreover, the exponential upper bound is obtained by a threshold test constructed from the joint Rényi projection, with threshold

$$\tau = \frac{\log(1/\varepsilon) - n(1-\lambda)\inf_{Q \in \mathcal{C}_1} \inf_{P \in \mathcal{C}_0} D_\lambda(Q \| P)}{\lambda}. \tag{7}$$

The proof of Theorem 2 is given in Appendix A-E.

Theorem 2 gives the finite sample achievability bound (6) without assuming that the projected pair is least favourable.

C. Phase transition under an exponential Type I constraint

We now specialise the finite sample bounds to a Type I error probability constraint that decays exponentially with sample size. For $r > 0$, set $\varepsilon = e^{-nr}$.

Theorem 3: Fix $r > 0$ and assume that

$$\lim_{\lambda\uparrow 1} D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = \lim_{\lambda\downarrow 1} D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = D(\mathcal{C}_1\|\mathcal{C}_0).$$

Suppose moreover that there exists a sequence $\lambda_k \uparrow 1$, with $\lambda_k \in (0,1)$, such that the assumptions of Theorem 2 hold at every λ_k. Then

$$\lim_{n\to\infty} \beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) = \begin{cases} 0, & 0 < r < D(\mathcal{C}_1\|\mathcal{C}_0), \\ 1, & r > D(\mathcal{C}_1\|\mathcal{C}_0). \end{cases} \tag{8}$$

No assertion is made when $r = D(\mathcal{C}_1\|\mathcal{C}_0)$.

The proof of Theorem 3 is given in Appendix A-F.

The order one limits assumed in Theorem 3 do not follow directly from the corresponding fixed pair result because the pair minimising the Rényi divergence may vary with the order. The following lemma gives a sufficient condition for convergence from above, while Corollary 6 (Appendix A-H) gives one for convergence from below.

Lemma 2: Assume that $D(\mathcal{C}_1\|\mathcal{C}_0) < +\infty$ and that, for every $\eta > 0$, there exist $P_\eta \in \mathcal{C}_0$, $Q_\eta \in \mathcal{C}_1$, and $\delta_\eta > 0$ such that $D(Q_\eta\|P_\eta) \le D(\mathcal{C}_1\|\mathcal{C}_0) + \eta$ and $D_{1+\delta_\eta}(Q_\eta\|P_\eta) < +\infty$. Then

$$\lim_{\lambda\downarrow 1} D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = D(\mathcal{C}_1\|\mathcal{C}_0).$$

The proof of Lemma 2 is given in Appendix A-G.

For simple binary testing, the finite sample achievability and converse bounds involve Rényi orders below and above one, respectively [6]. Their limits as the order approaches one recover the KL divergence and hence the phase transition threshold. In the composite setting, the same mechanism applies only after establishing the corresponding order-one limits for $D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)$, since the pair minimising the Rényi divergence may depend on λ.

We now specialise the result to compact convex classes with full support on a finite alphabet and characterise the exact error exponents on both sides of the phase transition.

IV. Exact error exponents on finite alphabets

Let $\mathcal{X} = \{1,\ldots,d\}$ be a finite alphabet and let $\Delta_d := \left\{p \in [0,1]^d \,\middle|\, \sum_{x=1}^d p(x) = 1\right\}$ denote the probability simplex on $\mathcal{X}$. Throughout this section, $\mathcal{C}_0, \mathcal{C}_1 \subseteq \Delta_d$ are nonempty, compact, and convex, and every distribution in $\mathcal{C}_0 \cup \mathcal{C}_1$ has full support.

Compactness and full support give a common strictly positive lower bound on the probability assigned to every point in the alphabet by every distribution in the two classes. Thus all distributions are mutually absolutely continuous, every pair has a strictly positive Hellinger integral, and the support conditions in Theorem 2 are automatic. Consequently, for every $\lambda \in (0,1)$ a projected pair $(Q^\star_\lambda, P^\star_\lambda)$ exists, with $h^\star_\lambda(x) = \log \frac{Q^\star_\lambda(x)}{P^\star_\lambda(x)}$, and Theorem 2 applies without further support assumptions.

Under the finite alphabet assumptions, both order one limits required by Theorem 3 hold. Convergence from below follows from Corollary 6 (Appendix A-H). For convergence from above, uniform full support ensures that the Rényi divergence of every finite order is finite for every pair, so Lemma 2 applies. Together with the achievability conditions established above, Theorem 3 therefore identifies $D(\mathcal{C}_1\|\mathcal{C}_0)$ as the phase transition threshold.

A. *Exact exponent in the achievable regime*

In the achievable regime $0 < r < D(\mathcal{C}_1\|\mathcal{C}_0)$, the projected test gives an achievable Type II error exponent. Every test admissible for the composite problem is also admissible for each simple binary problem obtained by selecting one distribution from each class. The composite exponent therefore cannot exceed the exponent for any fixed pair. To establish optimality, it is enough to show that the projected achievability exponent coincides with the smallest exponent over all such pairs.

Under the finite alphabet assumptions, the optimisation over the Rényi order and the pair of distributions has the required convexity-concavity structure for Sion's minimax Theorem [19, Th. 3.4], which allows the two optimisations to be interchanged. Compactness then gives attainment of both extrema. This identifies an order and a projected pair for which the composite exponent coincides with the corresponding simple binary exponent.

Theorem 4: Fix $0 < r < D(\mathcal{C}_1\|\mathcal{C}_0)$. Then the exact Type II error exponent exists and satisfies

$$\begin{aligned}\lim_{n\to\infty} -\frac{1}{n}\log \beta^\star_n(e^{-nr};\mathcal{C}_0,\mathcal{C}_1) &= \max_{0<\lambda<1} \frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) - r\right] \\ &= \min_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}} \max_{0<\lambda<1} \frac{1-\lambda}{\lambda}\left[D_\lambda(Q\|P) - r\right]. \end{aligned} \tag{9}$$

Moreover, there exist $\lambda^\star_r \in (0,1)$,$P^\star_r \in \mathcal{C}_0$, and $Q^\star_r \in \mathcal{C}_1$ such that $(Q^\star_r, P^\star_r)$ attains $D_{\lambda^\star_r}(\mathcal{C}_1\|\mathcal{C}_0)$ and $\lambda^\star_r$ attains the inner maximum for $(P^\star_r, Q^\star_r)$. The common value is

$$\frac{1-\lambda^\star_r}{\lambda^\star_r}\left[D_{\lambda^\star_r}(Q^\star_r\|P^\star_r) - r\right].$$

Equation (9) shows that, among all simple pairs $(P,Q) \in \mathcal{C}_0 \times \mathcal{C}_1$, the pair $(P^\star_r, Q^\star_r)$ attains the smallest Type II error exponent. The latter, is also the exponent of the composite problem, so $(P^\star_r, Q^\star_r)$ determines the decay rate of the optimal composite Type II error. In other words, $(P^\star_r, Q^\star_r)$ is least favourable at the level of the error exponent.

Corollary 1: For every $0 < r < D(\mathcal{C}_1\|\mathcal{C}_0)$, the Rényi order $\lambda^\star_r$ in Theorem 4 is unique. Moreover, the exact Type II error exponent is continuously differentiable with respect to r, with derivative: $-(1-\lambda^\star_r)/\lambda^\star_r$.
The proof is given in Appendix B-C.

Corollary 1 shows that the exact achievable exponent is attained at a unique Rényi order, and that this order can be recovered directly from the derivative of the exponent.

B. *Exact exponent in the converse regime*

We now consider $r > D(\mathcal{C}_1\|\mathcal{C}_0)$, where Theorem 3 shows that the optimal Type II error converges to one. The nontrivial quantity is $1 - \beta^\star_n(e^{-nr};\mathcal{C}_0,\mathcal{C}_1)$. The finite sample converse in Theorem 1 provides an exponential upper bound on this quantity. The following theorem determines its decay exponent and shows that the exponent in the converse is attained.

Theorem 5: For every $r > D(\mathcal{C}_1\|\mathcal{C}_0)$, the exact exponent of the residual probability exists and satisfies

$$\lim_{n\to\infty} -\frac{1}{n}\log\left(1-\beta_n^\star(e^{-nr};\mathcal{C}_0,\mathcal{C}_1)\right) = \sup_{\lambda>1}\frac{\lambda-1}{\lambda}\left[r - D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)\right]_+ \tag{10}$$

$$= \sup_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}}\sup_{\lambda>1}\frac{\lambda-1}{\lambda}\left[r - D_\lambda(Q\|P)\right]_+ . \tag{11}$$

Consequently,

$$1-\beta_n^\star(e^{-nr};\mathcal{C}_0,\mathcal{C}_1) = \exp\left\{-n\sup_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}}\sup_{\lambda>1}\frac{\lambda-1}{\lambda}\left[r - D_\lambda(Q\|P)\right]_+ + o(n)\right\}. \tag{12}$$

The proof is given in Appendix B-D.

Theorem 5 completes the exponent level characterisation on both sides of the phase transition threshold $D(\mathcal{C}_1\|\mathcal{C}_0)$ identified in Theorem 3. In the achievable regime, the exact Type II error exponent is the smallest exponent among the corresponding simple binary problems. In the converse regime, the exponent governing convergence of the Type II error to one is the largest strong converse exponent among the simple pairs.

C. Fixed Type I constraint and zero rate limit

The exponentially decaying Type I constraint excludes the boundary case $r = 0$. We now consider this boundary by fixing $\varepsilon \in (0,1)$. For simple binary testing, the Chernoff-Stein lemma gives the KL divergence from the null to the alternative, reversing the direction that determines the threshold rate. In the projected achievability bound, the relevant Rényi orders approach zero as $n \to \infty$. We first determine the corresponding order zero limit.

Lemma 3: For every $P \in \mathcal{C}_0$, $Q \in \mathcal{C}_1$, and $\lambda \in (0,1)$,

$$D(P\|Q) - \frac{\lambda}{2}\left[\log\frac{1}{\min_{\substack{R\in\mathcal{C}_0\cup\mathcal{C}_1\\ x\in\mathcal{X}}} R(x)}\right]^2 \le \frac{1-\lambda}{\lambda}D_\lambda(Q\|P) \le D(P\|Q).$$

Consequently,

$$\lim_{\lambda\downarrow 0}\frac{1-\lambda}{\lambda}D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = D(\mathcal{C}_0\|\mathcal{C}_1). \tag{13}$$

The proof is given in Appendix B-E.

The fixed Type I exponent for compact convex classes on a finite alphabet is a known composite Chernoff-Stein result [17], [20]. We recover it here directly from the finite sample Rényi achievability bound. This also identifies the fixed Type I regime with the order-zero boundary of the Rényi expression used above.

Corollary 2: For every fixed $\varepsilon \in (0,1)$,

$$\lim_{n\to\infty} -\frac{1}{n}\log\beta_n^\star(\varepsilon;\mathcal{C}_0,\mathcal{C}_1) = D(\mathcal{C}_0\|\mathcal{C}_1). \tag{14}$$

If $D(\mathcal{C}_0\|\mathcal{C}_1) = 0$, then

$$\beta_n^\star(\varepsilon;\mathcal{C}_0,\mathcal{C}_1) = 1-\varepsilon \qquad \forall n \ge 1.$$

The proof is given in Appendix B-F.

The fixed Type I constraint and the exponentially decaying Type I constraint lead to opposite KL directions. The phase transition threshold is $D(\mathcal{C}_1\|\mathcal{C}_0)$, while under a fixed Type I constraint $\varepsilon \in (0,1)$ the Type II error exponent is $D(\mathcal{C}_0\|\mathcal{C}_1)$. The former arises from Rényi orders approaching one, while the latter arises from orders approaching zero. When $\varepsilon = e^{-nr}$, the achievable exponent converges to the fixed Type I exponent as $r \downarrow 0$.

Corollary 3: If $D(\mathcal{C}_0\|\mathcal{C}_1) > 0$, then $\lim_{r\downarrow 0} \max_{0<\lambda<1} \frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) - r\right] = D(\mathcal{C}_0\|\mathcal{C}_1)$.
The proof is given in Appendix B-G.

V. Refining the analysis

Throughout this section, we retain the finite alphabet assumptions of Section IV. We first keep the projected log-likelihood ratio Equation (15) fixed and optimise its threshold directly under an arbitrary composite Type I constraint $\varepsilon \in (0,1)$. We then return to the exponentially decaying constraint $\varepsilon = e^{-nr}$ in the achievable regime $0 < r < D(\mathcal{C}_1\|\mathcal{C}_0)$, where Theorem 4 identifies the exact Type II error exponent. In this regime, we determine the polynomial order of the optimal Type II error and the corresponding logarithmic correction to the threshold in (7) in Theorem 2.

The preceding results are sufficient to recover the correct exponential behaviour of the optimal Type II error and the phase transition between the achievable and converse regimes. An advantage of the achievability bound in Theorem 2 is that it is explicit and amenable to analytical study. It can, however, be tightened at finite sample size at the cost of a less explicit construction that, in general, requires numerical evaluation.

A. Tightening the achievability bound

The threshold in (7) is chosen using an exponential upper bound on the Type I error so that the constraint holds uniformly for every $P \in \mathcal{C}_0$. Consequently, the resulting Type I error may be strictly smaller than ε. A tighter bound is obtained by keeping the log-likelihood ratio selected by the Rényi projection fixed and calibrating its threshold directly against the composite Type I constraint.

Proposition 1: Fix $\lambda \in (0,1)$ and a projected pair $(Q_\lambda^\star, P_\lambda^\star)$ satisfying Theorem 2, and define

$$h_\lambda^\star(x) := \log \frac{Q_\lambda^\star(x)}{P_\lambda^\star(x)}. \tag{15}$$

For each $n \geq 1$, $\tau \in \mathbb{R}$, and $\eta \in [0,1]$, define

$$\psi_{\tau,\eta}^\star(x^n) := \mathbb{1}\left\{\sum_{i=1}^n h_\lambda^\star(x_i) > \tau\right\} + \eta \mathbb{1}\left\{\sum_{i=1}^n h_\lambda^\star(x_i) = \tau\right\}.$$

Then, for every $n \geq 1$ and $\varepsilon \in (0,1)$, there exist $\widehat{\tau} \in \mathbb{R}$ and $\widehat{\eta} \in [0,1]$ such that $\sup_{P\in\mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}\left[\psi_{\widehat{\tau},\widehat{\eta}}^\star\right] = \varepsilon$. The induced test uniquely minimises the Type II error over all tests $\psi_{\tau,\eta}^\star$ satisfying the Type I constraint. Moreover,

$$\beta_n^\star(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) \leq \sup_{Q\in\mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}\left[1 - \psi_{\widehat{\tau},\widehat{\eta}}^\star\right]. \tag{16}$$

The proof of Proposition 1 is given in Appendix C-A.

For a fixed Rényi order, Proposition 1 gives the smallest Type II upper bound obtainable from threshold tests based on the corresponding projected log-likelihood ratio Equation (15). In particular, it cannot be worse than the explicit threshold choice in Theorem 2. The bound may be tightened further by optimising over the Rényi order, but the projected pair and calibrated threshold then both depend on the order. This optimisation is generally less amenable to analytical treatment, and in Section VII we evaluate it numerically.

The uniqueness in Proposition 1 concerns the induced test function, not necessarily its representation by (τ, η).Different pairs (τ, η) can represent the same boundary rule.For singleton classes, Appendix C-B shows that $h^\star_\lambda$ is the ordinary log-likelihood ratio and the optimised threshold test is the NP test. For general composite classes, the optimisation remains restricted to threshold tests based on a projected log-likelihood ratio Equation (15) and is not asserted to solve the unrestricted composite problem.

B. Polynomial refinement

Theorem 4 determines the exponential rate, while Proposition 1 shows that the projected threshold may be improved when the projected log-likelihood ratio Equation (15) (equation (15)) is held fixed. We now determine the behaviour beyond the exponential scale at the selected order $\lambda^\star_r$. Under the exponential constraint $\varepsilon = e^{-nr}$, the threshold (τ) given by Theorem 2 at the selected order is

$$n\left[r - \frac{1-\lambda^\star_r}{\lambda^\star_r}\left(D_{\lambda^\star_r}(Q^\star_r \| P^\star_r) - r\right)\right].$$

A sharper analysis of the Type I error at reveals an additional factor of order $n^{-1/2}$. The projected threshold (τ) therefore leaves part of the available Type I error unused. Recovering it requires a correction of order $\log n$ and produces a polynomial factor in the Type II error.

Proposition 2: Fix $0 < r < D(\mathcal{C}_1 \| \mathcal{C}_0)$, and let $(\lambda^\star_r, P^\star_r, Q^\star_r)$ satisfy Theorem 4. Then

$$\beta^\star_n(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) = \Theta\left(n^{-1/(2\lambda^\star_r)} \exp\left\{-n\frac{1-\lambda^\star_r}{\lambda^\star_r}\left[D_{\lambda^\star_r}(Q^\star_r \| P^\star_r) - r\right]\right\}\right). \tag{17}$$

Moreover, the matching upper bound is achieved by a projected log-likelihood ratio Equation (15) test with threshold

$$n\left[r - \frac{1-\lambda^\star_r}{\lambda^\star_r}\left(D_{\lambda^\star_r}(Q^\star_r \| P^\star_r) - r\right)\right] - \frac{\log n}{2\lambda^\star_r} + O(1).$$

The proof is given in Appendix C-D.

Equation (17) determines the polynomial order but does not identify the leading multiplicative constant under the assumptions of Proposition 2. Stronger constant level conclusions require additional regularity conditions.

Corollary 4: For every $0 < r < D(\mathcal{C}_1 \| \mathcal{C}_0)$, the polynomial power in (17) satisfies

$$\frac{1}{2\lambda^\star_r} = \frac{1}{2}\left[1 - \frac{d}{dr}\max_{0<\lambda<1}\frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) - r\right]\right]. \tag{18}$$

The proof of Corollary 4 is given in Appendix C-E.

Corollary 4 shows that the polynomial power is determined directly by the slope of the exact Type II error exponent. The associated projected pair $(Q^\star_r, P^\star_r)$, however, need not be unique.

VI. When is the projected pair least favourable?

Classical robust testing results establish least favourability under structural conditions such as stochastic ordering or related dominance properties of the likelihood ratio [8]–[10]. Under such conditions, a particular pair of distributions attains the suprema defining the Type I and Type II errors, reducing the composite problem to a simple binary test.

The Rényi projection used in the preceding sections is selected because it provides uniform exponential control over the two classes. This property alone does not imply that the projected pair is least favourable at finite sample size. Gül [11] gives convex uncertainty classes for which a single pair is optimal for every Rényi order in $(0, 1)$ but fails the stronger convex order condition required for finite sample least favourability.

We therefore ask under what additional conditions the projected pair also attains the composite error probabilities. Fix $\lambda \in (0, 1)$ and a projected pair $(Q^\star_\lambda, P^\star_\lambda)$, and let $\psi^\star_{\tau,\eta}$ denote the projected threshold test introduced in Section V-A. The following condition imposes stochastic ordering directly on the projected log-likelihood ratio Equation (15).

Proposition 3: Suppose that, for every $t \in \mathbb{R}$, $\forall P \in \mathcal{C}_0$, and $\forall Q \in \mathcal{C}_1$,

$$P(h^\star_\lambda \geq t) \leq P^\star_\lambda(h^\star_\lambda \geq t), \quad \text{and} \quad Q(h^\star_\lambda \leq t) \leq Q^\star_\lambda(h^\star_\lambda \leq t).$$

Then, for every $n \geq 1$, $\tau \in \mathbb{R}$, and $\eta \in [0, 1]$,

$$\sup_{P \in \mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}[\psi^\star_{\tau,\eta}] = \mathbb{E}_{(P^\star_\lambda)^{\otimes n}}[\psi^\star_{\tau,\eta}], \quad \text{and} \quad \sup_{Q \in \mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}[1 - \psi^\star_{\tau,\eta}] = \mathbb{E}_{(Q^\star_\lambda)^{\otimes n}}[1 - \psi^\star_{\tau,\eta}].$$

The proof is given in Appendix D-A.

Thus, under the stochastic ordering conditions of Proposition 3, the projected pair attains both composite error suprema for every projected threshold test. To obtain an exact reduction of the unrestricted composite problem, the same threshold test must also be optimal for the projected simple pair.

Corollary 5: Assume the conditions of Proposition 3. Fix $n \geq 1$ and $\varepsilon \in (0, 1)$. Suppose that $\psi^\star_{\tau,\eta}$ is optimal for testing $(P^\star_\lambda)^{\otimes n}$ against $(Q^\star_\lambda)^{\otimes n}$ under the constraint $\mathbb{E}_{(P^\star_\lambda)^{\otimes n}}[\psi^\star_{\tau,\eta}] \leq \varepsilon$. Then

$$\beta^\star_n(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) = \beta^\star_n(\varepsilon; P^\star_\lambda, Q^\star_\lambda).$$

The proof is given in Appendix D-B.

The stochastic ordering conditions in Proposition 3 hold for separated one parameter natural exponential families.

Definition 4: Let $\{P_\theta : \theta \in \Theta\}$ have common support and densities $p_\theta(x) = h_0(x) \exp\{\theta T(x) - \psi(\theta)\}$, and $\theta \in \Theta$, where $\Theta \subseteq \mathbb{R}$ is an open interval, $h_0 > 0$, and ψ is finite, differentiable, and strictly convex. Let $\mathcal{C}_0$ and $\mathcal{C}_1$ consist of P_θ over two disjoint closed intervals $[\theta^{\mathcal{C}_0}_-, \theta^{\mathcal{C}_0}_+]$ and $[\theta^{\mathcal{C}_1}_-, \theta^{\mathcal{C}_1}_+]$, respectively.

Proposition 4: For every $n \geq 1$ and $\varepsilon \in (0, 1)$,

$$\beta^\star_n(\varepsilon; \mathcal{C}_0, \mathcal{C}_1) = \begin{cases} \beta^\star_n\left(\varepsilon; P_{\theta^{\mathcal{C}_0}_+}, P_{\theta^{\mathcal{C}_1}_-}\right), & \theta^{\mathcal{C}_0}_+ < \theta^{\mathcal{C}_1}_-, \\ \beta^\star_n\left(\varepsilon; P_{\theta^{\mathcal{C}_0}_-}, P_{\theta^{\mathcal{C}_1}_+}\right), & \theta^{\mathcal{C}_1}_+ < \theta^{\mathcal{C}_0}_-. \end{cases}$$

Moreover, for every $\lambda \in (0, 1)$, the corresponding endpoint pairs a joint Rényi projection.

The proof is given in Appendix D-D.

Hence, for separated one parameter natural exponential families, the same endpoint pair is selected by the Rényi projection and is least favourable at every sample size.

VII. Numerical illustrations

Beyond characterising the phase transition threshold Equation (8), and the corresponding error exponents, we illustrate the finite sample behaviour of the bounds derived above in three settings.

First, we consider nonordered affine classes of ternary distributions.

Second, we consider separated one parameter families, including Bernoulli, Poisson, Gaussian, and exponential models, for which Proposition 4 gives an exact reduction to a simple binary testing problem.

Third, we consider nonordered affine ternary classes under fixed and subexponentially decreasing Type I constraints, with $\varepsilon_n = 0.01$ and $\varepsilon_n = 1/n$. In each setting, we compare the relevant Rényi bounds with the optimal Type II error.

Further details of the numerical calculations, including the evaluation of the composite Rényi divergences, the computation of the phase transition threshold, and the optimisation over the Rényi order, are provided in Appendix E.

The first setting considers two affine classes of ternary distributions, for $s, t \in [0, 1]$

$$P_s = (1-s)(0.327, 0.418, 0.255) + s(0.563, 0.266, 0.171),$$

$$Q_t = (1-t)(0.143, 0.357, 0.500) + t(0.379, 0.205, 0.416).$$

These classes are compact, convex, and have uniform full support, with $D(\mathcal{C}_1 \| \mathcal{C}_0) = 0.094$; numerically.

For the achievable regime, we take $r = 0.35D(\mathcal{C}_1 \| \mathcal{C}_0) = 0.033$. For each sample size n, we compare the optimal Type II error with the achievability bound obtained by calibrating the threshold and boundary randomisation of the projected log-likelihood ratio Equation (15) directly under the Type I constraint $\varepsilon = e^{-nr}$. The left panel of Fig. 1 shows this comparison.

The oscillations visible in the achievability curve at small and moderate sample sizes arise from the finite number of values taken by the projected log-likelihood ratio Equation (15) at each sample size n. Consequently, the admissible threshold values and the corresponding boundary randomisation change discretely with n.

For the converse regime, we take $r = 1.5D(\mathcal{C}_1 \| \mathcal{C}_0) = 0.142$. The right panel of Fig. 1 compares the Rényi converse in Theorem 1 with the optimal Type II error. For reference, we also include a Fano-style converse bound. The Rényi converse is stronger over the displayed sample sizes and captures the convergence of the Type II error towards one, whereas the Fano-style converse bound remains separated from one.

The second setting considers separated one parameter Bernoulli, Poisson, Gaussian, and exponential families covered by Proposition 4, under the exponentially decaying Type I constraint $\varepsilon = e^{-nr}$. For each family, we show the achievable regime at $r = 0.75r_{\mathrm{c}}$ and the converse regime at $r = 1.90r_{\mathrm{c}}$.

In the achievable regime, the endpoint pair selected by the Rényi projection is also least favourable by Proposition 4, so the composite problem reduces exactly to the corresponding simple binary problem. The optimised projected threshold test therefore coincides with the NP test, and its Type II error equals the exact composite Type II error. This agreement is shown in the left columns of Figs. 2 and 3.

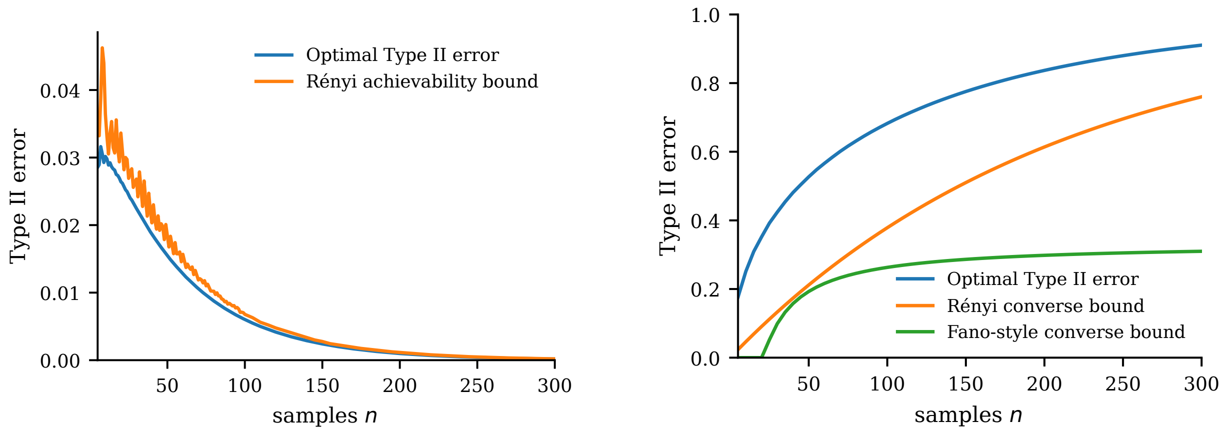

Fig. 1. Finite sample Type II error and Rényi bounds for the nonordered affine ternary classes under $\varepsilon = e^{-nr}$. The left panel considers $r = 0.35D(\mathcal{C}_1\|\mathcal{C}_0)$ and the right panel considers $r = 1.5D(\mathcal{C}_1\|\mathcal{C}_0)$.

The converse regime behaves differently. Although Proposition 4 still reduces the composite problem exactly to the corresponding simple binary problem, the finite sample Rényi converse gives a lower bound on the optimal Type II error rather than the exact NP performance [6]. The curves are therefore not expected to coincide at finite sample size. Nevertheless,the Rényi converse remains tight over the displayed finite sample range and closely tracks the Type II error as it approaches one,consistent withTheorem 5. This behaviour is shown in the right columns ofFigs. 2 and 3. Consistent with the phase transition in Theorem 3, the Type II error decreases towards zero in the left columns and approaches one in the right columns of Figs. 2 and 3.

The final setting considers composite testing when the Type I error constraint is fixed ($\varepsilon_n = 0.01$) and when it decreases subexponentially with sample size ($\varepsilon_n = 1/n$).

For $s, t \in [0, 1]$, let

$$\widetilde{P}_s = (1-s)(0.33, 0.33, 0.34) + s(0.33, 0.35, 0.32),$$
$$\widetilde{Q}_t = (1-t)(0.203, 0.428, 0.369) + t(0.370, 0.455, 0.175).$$

Here, $\widetilde{\mathcal{C}}_0 = \{\widetilde{P}_s : s \in [0,1]\}$ and $\widetilde{\mathcal{C}}_1 = \{\widetilde{Q}_t : t \in [0,1]\}$. These classes are compact, convex, and have uniform full support, with $D(\widetilde{\mathcal{C}}_1\|\widetilde{\mathcal{C}}_0) = 0.017$ and $D(\widetilde{\mathcal{C}}_0\|\widetilde{\mathcal{C}}_1) = 0.016$; numerically.

For achievability, we use the calibrated projected threshold test of Proposition 1. The left column of Fig. 4 compares this bound with the optimal Type II error. The top panel uses the fixed constraint $\varepsilon_n = 0.01$, while the bottom panel uses the subexponential constraint $\varepsilon_n = 1/n$.

For the converse, we use Theorem 1 together with a Fano-style converse bound. The right column of Fig. 4 compares these bounds with the numerical Type II error under each Type I constraint. In both cases, the Rényi converse is tighter and remains positive over a larger range of sample sizes. Both converse bounds eventually decrease to zero because $\log(1/\varepsilon_n) = o(n)$.

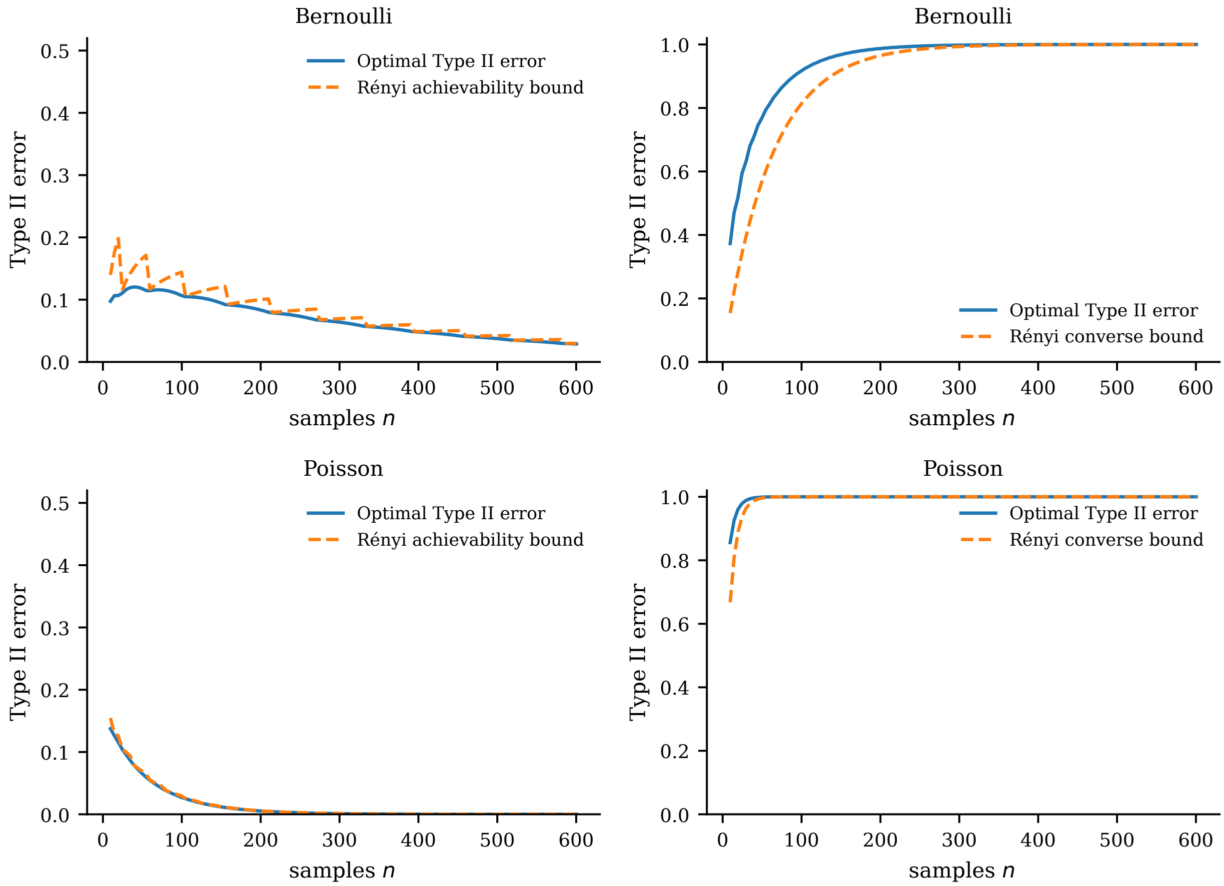

Fig. 2. Finite sample Type II error for the separated Bernoulli and Poisson families. The left column shows the achievable regime at $r = 0.75r_{\mathrm{c}}$, comparing the optimal Type II error with the Rényi achievability bound. The right column shows the converse regime at $r = 1.90r_{\mathrm{c}}$, comparing the optimal Type II error with the Rényi converse bound.

VIII. Conclusion

We derived finite sample Rényi achievability and converse bounds for composite binary hypothesis testing under a uniform Type I constraint (ε). The bounds identify the phase transition under $\varepsilon = e^{-nr}$ and, for compact convex classes with full support on a finite alphabet, give the exact exponents on both sides of the transition, including the strong converse exponent above the phase transition threshold. A central ingredient of the achievability result is the joint Rényi projection, whose likelihood ratio provides uniform control over both classes without requiring the projected pair to be least favourable at finite sample size. The same Rényi structure recovers the fixed Type I composite Chernoff–Stein exponent through the order zero limit and determines the polynomial dependence on sample size at the selected order. Finally, we identified additional stochastic ordering conditions under which the projected pair is also least favourable, giving an exact finite sample reduction for separated one parameter natural exponential families.

Several questions remain open. For instance, the calibrated achievability bound currently requires numerical opti-

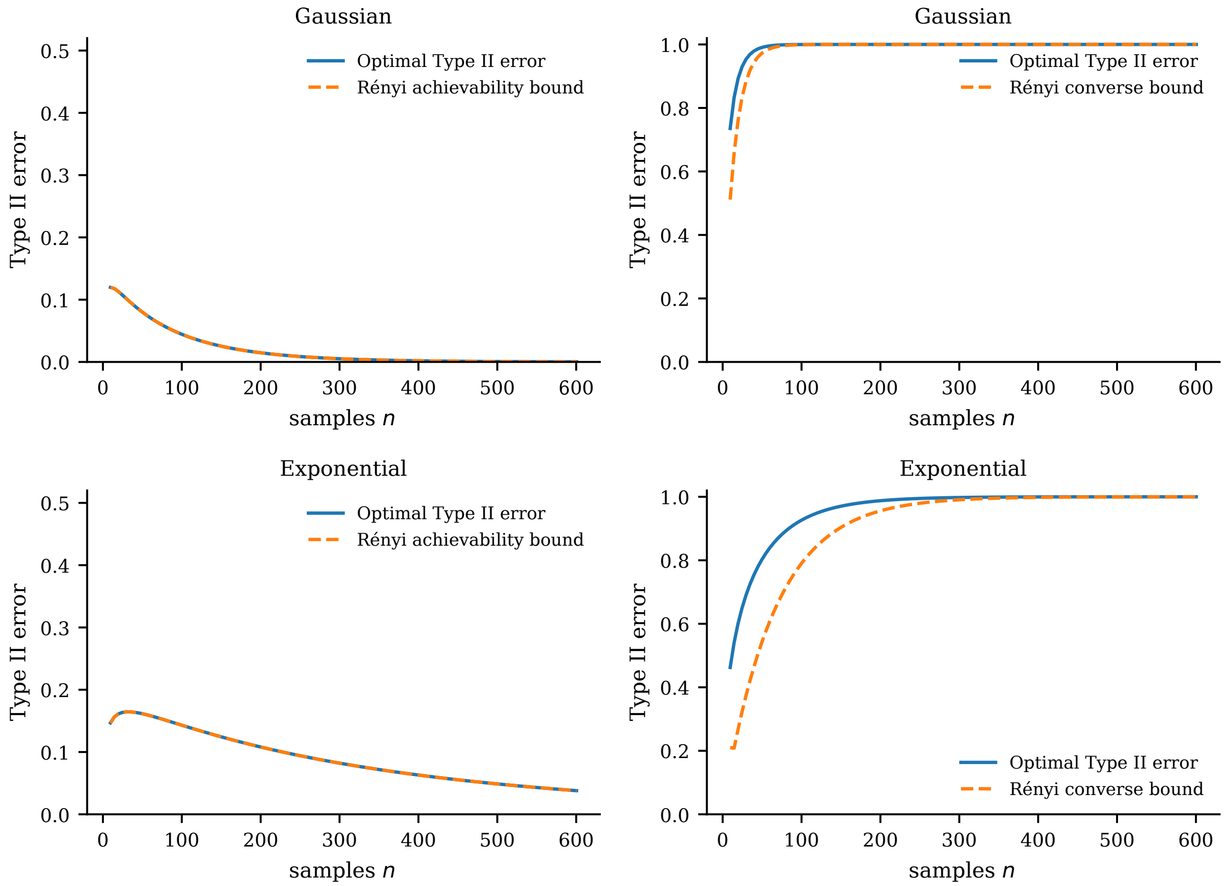

Fig. 3. Finite sample Type II error for the separated Gaussian and exponential families. The left column shows the achievable regime at $r = 0.75 r_{\mathrm{c}}$, comparing the optimal Type II error with the Rényi achievability bound. The right column shows the converse regime at $r = 1.90 r_{\mathrm{c}}$, comparing the optimal Type II error with the Rényi converse bound.

misation. It remains to be seen whether the tightened bound admits an analytical characterisation and, consequently, a closed form expression. Our polynomial refinement provides a sharper description of its finite sample behaviour by determining the power of n and the logarithmic threshold correction, but does not identify the leading multiplicative constant. On the converse side, our bound applies the finite sample Rényi converse to individual pairs $(P^{\otimes n}, Q^{\otimes n})$. One may instead consider Bayesian mixtures such as $\int P^{\otimes n} d\Pi_0(P)$ and $\int Q^{\otimes n} d\Pi_1(Q)$, each of which gives a valid lower bound on the composite minimax error. Whether optimising the Rényi converse over such mixture pairs yields tighter finite sample bounds remains open. Since the leading exponential rate is already exact under the finite alphabet assumptions, any improvement must occur at finer finite sample scales.

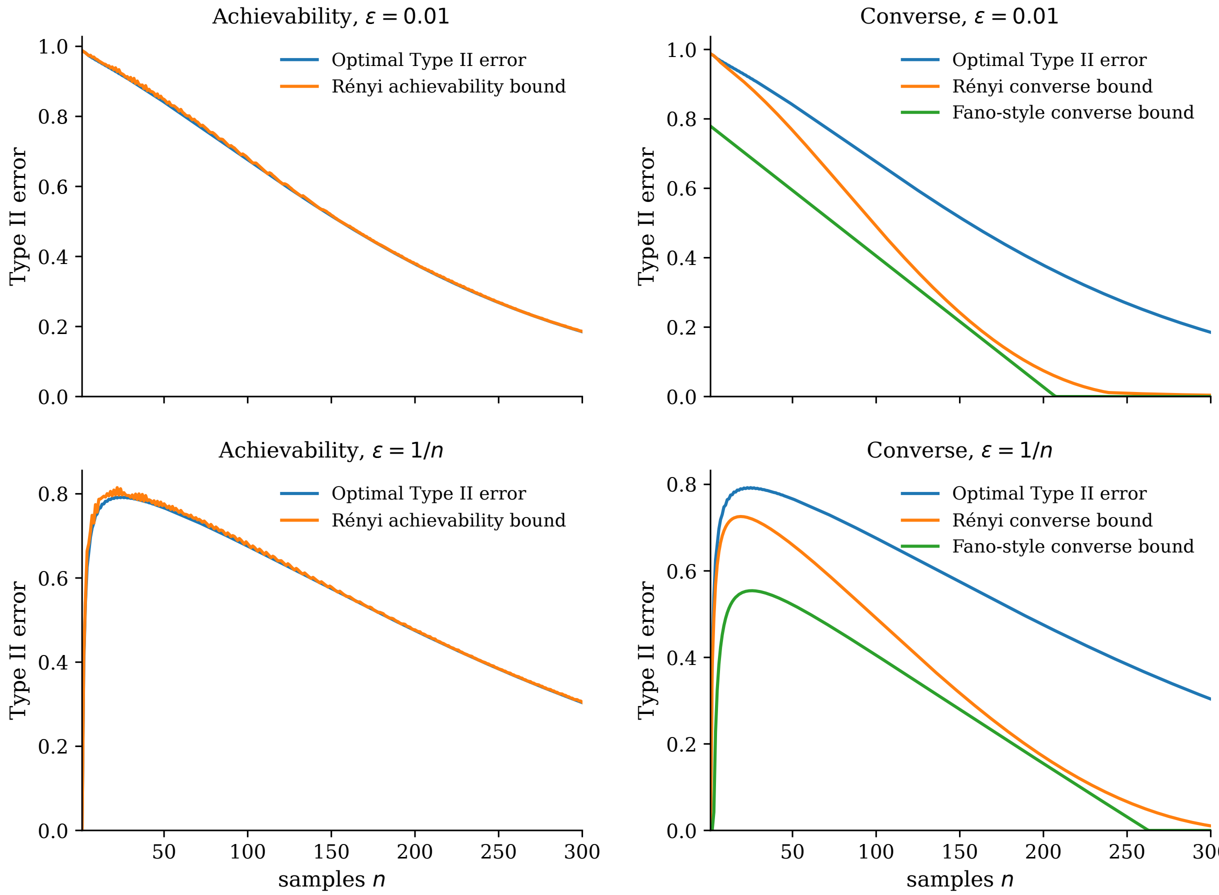

Fig. 4. Finite sample Type II error and bounds for the nonordered affine ternary classes under fixed and subexponential Type I constraints. The top row uses $\varepsilon_n = 0.01$ and the bottom row uses $\varepsilon_n = 1/n$. The left column shows the calibrated Rényi achievability bound and the right column shows the Rényi and the Fano-style converse bounds.

Appendix A
Bounds

A. Proof of Theorem 1

Proof: Fix an admissible test φ_n, $P \in \mathcal{C}_0$, $Q \in \mathcal{C}_1$, and $\lambda > 1$. Suppose first that $D_\lambda(Q\|P) < +\infty$. Then $Q \ll P$, and Hölder's inequality, the Type I constraint, and additivity of the Rényi divergence give

$$\mathbb{E}_{Q^{\otimes n}}[\varphi_n] = \mathbb{E}_{P^{\otimes n}}\left[\frac{dQ^{\otimes n}}{dP^{\otimes n}}\varphi_n\right] \leq \exp\left\{-\frac{\lambda-1}{\lambda}\left[\log\frac{1}{\varepsilon} - nD_\lambda(Q\|P)\right]\right\},$$

where $\varphi_n^{\lambda/(\lambda-1)} \leq \varphi_n$. Combining this with $\mathbb{E}_{Q^{\otimes n}}[\varphi_n] \leq 1$ gives

$$\mathbb{E}_{Q^{\otimes n}}[1-\varphi_n] \geq 1 - \exp\left\{-\frac{\lambda-1}{\lambda}\left[\log\frac{1}{\varepsilon} - nD_\lambda(Q\|P)\right]_+\right\}.$$

For $D_\lambda(Q\|P) = +\infty$, the bound is trivial. If $D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = +\infty$, the resulting class bound is trivial. Otherwise, choose $(Q_k, P_k) \in \mathcal{C}_1 \times \mathcal{C}_0$ such that $D_\lambda(Q_k\|P_k) \longrightarrow D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)$. Since the preceding inequality holds for every pair, letting $k \to \infty$ gives

$$\sup_{Q \in \mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}[1-\varphi_n] \geq 1 - \exp\left\{ -\frac{\lambda-1}{\lambda} \left[\log\frac{1}{\varepsilon} - nD_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) \right]_+ \right\}.$$

Since this holds for every $\lambda > 1$, taking the supremum over $\lambda > 1$ and then the infimum over all admissible tests yields (4). ■

B. *Proof of Lemma 1*

Proof: The moment assumptions imply $P(h = +\infty) = 0 \quad \forall P \in \mathcal{C}_0$, and $Q(h = -\infty) = 0 \quad \forall Q \in \mathcal{C}_1$, because the corresponding exponential would otherwise have infinite expectation. Let $S_n(x^n) := \sum_{i=1}^n h(x_i)$, with $S_n = 0$ on sequences containing both $+\infty$ and $-\infty$. This set is null under every relevant product law, so the convention does not affect either error probability. Independence gives

$$\sup_{P \in \mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}[e^{\lambda S_n}] \leq e^{n(\lambda-1)\xi_0}, \quad \text{and} \quad \sup_{Q \in \mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}[e^{(\lambda-1)S_n}] \leq e^{n(\lambda-1)\xi_1}.$$

Markov's inequality gives

$$\alpha_n(\psi_{n,\tau}; \mathcal{C}_0) \leq e^{-\lambda\tau + n(\lambda-1)\xi_0}.$$

Since $\lambda - 1 < 0$, $S_n < \tau$ is equivalent to $e^{(\lambda-1)S_n} > e^{(\lambda-1)\tau}$. Therefore

$$\beta_n(\psi_{n,\tau}; \mathcal{C}_1) = \sup_{Q \in \mathcal{C}_1} Q^{\otimes n}(S_n < \tau) \leq e^{(1-\lambda)\tau + n(\lambda-1)\xi_1}.$$

Finally, the Type I bound is at most ε whenever

$$\tau \geq \frac{\log(1/\varepsilon) - n(1-\lambda)\xi_0}{\lambda}.$$

■

C. *Lemma 4*

Lemma 4: Let $0 < \alpha < 1$. For probability densities $p, \widetilde{p}, q, \widetilde{q}$ with respect to μ,

$$\left| \int q^\alpha p^{1-\alpha}\, d\mu - \int \widetilde{q}^\alpha \widetilde{p}^{1-\alpha}\, d\mu \right| \leq \|q - \widetilde{q}\|_1^\alpha + \|p - \widetilde{p}\|_1^{1-\alpha}.$$

Hence H_α is continuous in the product $L^1(\mu)$ norm and weakly upper semicontinuous, while $(Q,P) \mapsto D_\alpha(Q\|P)$ is weakly lower semicontinuous.

Proof: Insert and subtract $\widetilde{q}^\alpha p^{1-\alpha}$. Using $|u^\alpha - v^\alpha| \leq |u-v|^\alpha$ and Hölder's inequality,

$$\left| \int q^\alpha p^{1-\alpha}\, d\mu - \int \widetilde{q}^\alpha \widetilde{p}^{1-\alpha}\, d\mu \right| \leq \int |q - \widetilde{q}|^\alpha p^{1-\alpha}\, d\mu + \int \widetilde{q}^\alpha |p - \widetilde{p}|^{1-\alpha}\, d\mu \leq \|q - \widetilde{q}\|_1^\alpha + \|p - \widetilde{p}\|_1^{1-\alpha}.$$

Thus H_α is norm continuous. Its superlevel sets are convex by joint concavity and norm closed by the displayed bound, hence weakly closed. Therefore H_α is weakly upper semicontinuous. Since $D_\alpha = -\frac{1}{1-\alpha}\log H_\alpha$, with $D_\alpha = +\infty$ when $H_\alpha = 0$, the divergence is weakly lower semicontinuous. ■

D. Lemma 5

The proof of Theorem 2 uses the following lemma, which allows the maximising function to vanish on sets of positive measure.

Lemma 5: Let $0 < \alpha < 1$, and let w, x, y be nonnegative measurable functions that are finite μ almost everywhere. For $y_t = (1-t)y + tx$, suppose that

$$0 < \int wy^\alpha \, d\mu < +\infty, \qquad \int wy_t^\alpha \, d\mu \le \int wy^\alpha \, d\mu \quad \forall t \in [0,1].$$

Then

$$\mu\{w > 0,\ y = 0,\ x > 0\} = 0$$

and

$$\int_{\{y>0\}} wy^{\alpha-1}x \, d\mu \le \int wy^\alpha \, d\mu.$$

Moreover,

$$\lim_{t\downarrow 0} \frac{\int wy_t^\alpha \, d\mu - \int wy^\alpha \, d\mu}{t} = \alpha \left[\int_{\{y>0\}} wy^{\alpha-1}x \, d\mu - \int wy^\alpha \, d\mu \right] \le 0. \tag{19}$$

Proof: Set $I = \int wy^\alpha \, d\mu, \quad S = \{y > 0\}, \quad A = \{w > 0,\ y = 0\}$. For $t > 0$,

$$I \ge \int wy_t^\alpha \, d\mu \ge (1-t)^\alpha I + t^\alpha \int_A wx^\alpha \, d\mu,$$

and hence

$$0 \le \int_A wx^\alpha \, d\mu \le I \frac{1-(1-t)^\alpha}{t^\alpha} \le It^{1-\alpha}.$$

Letting $t \downarrow 0$ gives the support assertion. For $r \ge 0$, let

$$g_t(r) = \frac{(1-t+tr)^\alpha - 1}{t}.$$

Then

$$\int_S wy^\alpha g_t(x/y) \, d\mu \le 0, \quad g_t(r) \ge -1, \quad g_t(r) \to \alpha(r-1).$$

Fatou's lemma gives

$$(1-\alpha)I + \alpha \int_S wy^{\alpha-1}x \, d\mu \le I,$$

which proves the integrability bound. Concavity gives $g_t(r) \le \alpha(r-1)$, so $|g_t(r)| \le 1+\alpha r$. Dominated convergence then yields (19). ■

E. Proof of Theorem 2

Proof: By Lemma 4, the Hellinger integral Equation (2) attains its maximum on the weakly compact product of the two density classes. Since $\lambda - 1 < 0$, every maximising pair is a joint Rényi projection. Write

$$p_\lambda^\star = \frac{dP_\lambda^\star}{d\mu}, \qquad q_\lambda^\star = \frac{dQ_\lambda^\star}{d\mu},$$

and choose measurable versions

$$a_\lambda^\star = \frac{dP_\lambda^\star}{d(P_\lambda^\star + Q_\lambda^\star)}, \qquad b_\lambda^\star = \frac{dQ_\lambda^\star}{d(P_\lambda^\star + Q_\lambda^\star)}.$$

Define

$$h_\lambda^\star(x) = \begin{cases} \log(b_\lambda^\star(x)/a_\lambda^\star(x)), & a_\lambda^\star(x) > 0,\ b_\lambda^\star(x) > 0, \\ -\infty, & a_\lambda^\star(x) > 0,\ b_\lambda^\star(x) = 0, \\ +\infty, & a_\lambda^\star(x) = 0,\ b_\lambda^\star(x) > 0, \\ 0, & a_\lambda^\star(x) = b_\lambda^\star(x) = 0. \end{cases}$$

Fix $P \in \mathcal{C}_0$ and $Q \in \mathcal{C}_1$, with densities p and q. Optimality of the projected pair and convexity give, for every $t \in [0,1]$,

$$\int ((1-t)q_\lambda^\star + tq)^\lambda (p_\lambda^\star)^{1-\lambda}\, d\mu \le H_\lambda(Q_\lambda^\star, P_\lambda^\star), \qquad \int (q_\lambda^\star)^\lambda ((1-t)p_\lambda^\star + tp)^{1-\lambda}\, d\mu \quad \le H_\lambda(Q_\lambda^\star, P_\lambda^\star).$$

Since $H_\lambda(Q_\lambda^\star, P_\lambda^\star) \in (0,1]$, Lemma 5 yields

$$Q\{p_\lambda^\star > 0,\ q_\lambda^\star = 0\} = 0, \qquad\qquad P\{p_\lambda^\star = 0,\ q_\lambda^\star > 0\} = 0,$$

$$\int_{\{p_\lambda^\star q_\lambda^\star > 0\}} q (q_\lambda^\star)^{\lambda-1} (p_\lambda^\star)^{1-\lambda}\, d\mu \le H_\lambda(Q_\lambda^\star, P_\lambda^\star), \quad \int_{\{p_\lambda^\star q_\lambda^\star > 0\}} p (q_\lambda^\star)^{\lambda} (p_\lambda^\star)^{-\lambda}\, d\mu \le H_\lambda(Q_\lambda^\star, P_\lambda^\star).$$

The common zero set of $p_\lambda^\star$ and $q_\lambda^\star$ is null under every $R \in \mathcal{C}_0 \cup \mathcal{C}_1$ by assumption. Together with the preceding support relations, this gives

$$\mathbb{E}_P[e^{\lambda h_\lambda^\star}] \le H_\lambda(Q_\lambda^\star, P_\lambda^\star), \qquad \mathbb{E}_Q[e^{(\lambda-1)h_\lambda^\star}] \le H_\lambda(Q_\lambda^\star, P_\lambda^\star).$$

Therefore

$$\sup_{P \in \mathcal{C}_0} \mathbb{E}_P[e^{\lambda h_\lambda^\star}] \le e^{(\lambda-1)D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)}, \qquad \sup_{Q \in \mathcal{C}_1} \mathbb{E}_Q[e^{(\lambda-1)h_\lambda^\star}] \le e^{(\lambda-1)D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)}.$$

Apply Lemma 1 with $h = h_\lambda^\star$, $\xi_0 = \xi_1 = D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)$, and

$$\tau = \frac{\log(1/\varepsilon) - n(1-\lambda)D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)}{\lambda}.$$

The resulting test is admissible and satisfies

$$\beta_n(\psi_{n,\tau}; \mathcal{C}_1) \le \exp\left[-\frac{1-\lambda}{\lambda}\left(nD_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) - \log\frac{1}{\varepsilon}\right)\right].$$

The constant randomised test $\varphi_n \equiv \varepsilon$ has Type II error $1 - \varepsilon$. Taking the smaller of the two bounds yields (6). ■

F. Proof of Theorem 3

Proof: Set $\varepsilon = e^{-nr}$. If $0 < r < D(\mathcal{C}_1 \| \mathcal{C}_0)$, then $D_{\lambda_k}(\mathcal{C}_1 \| \mathcal{C}_0) \to D(\mathcal{C}_1 \| \mathcal{C}_0)$, so for some fixed k, $D_{\lambda_k}(\mathcal{C}_1 \| \mathcal{C}_0) > r$. By Theorem 2,

$$\beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \le \exp\left\{-n\frac{1-\lambda_k}{\lambda_k}\left[D_{\lambda_k}(\mathcal{C}_1 \| \mathcal{C}_0) - r\right]\right\}.$$

The exponent coefficient is strictly positive, hence $\beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \to 0$. If $r > D(\mathcal{C}_1 \| \mathcal{C}_0)$, the limit from above gives some $\lambda > 1$ such that $D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) < r$. By Theorem 1,

$$\beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \ge 1 - \exp\left\{-n\frac{\lambda-1}{\lambda}\left[r - D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)\right]\right\}.$$

Again the exponent coefficient is strictly positive, hence $\beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \to 1$. ■

G. *Proof of Lemma 2*

Proof: Monotonicity gives $D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) \geq D(\mathcal{C}_1\|\mathcal{C}_0) \quad \forall\lambda > 1$. Fix $\eta > 0$ and choose $(Q_\eta, P_\eta, \delta_\eta)$ as in the assumptions. Since $D_{1+\delta_\eta}(Q_\eta\|P_\eta) < +\infty$, the fixed pair Rényi to KL limit gives, for all $\lambda > 1$ sufficiently close to one,

$$D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) \leq D_\lambda(Q_\eta\|P_\eta) \leq D(Q_\eta\|P_\eta) + \eta \leq D(\mathcal{C}_1\|\mathcal{C}_0) + 2\eta.$$

Letting $\lambda \downarrow 1$ and then $\eta \downarrow 0$ proves the result. ■

H. *Corollary 6*

Corollary 6: Suppose that all laws in $\mathcal{C}_0$ and $\mathcal{C}_1$ admit densities with respect to a common σ-finite measure μ, and that, when identified with their densities with respect to μ, $\mathcal{C}_0$ and $\mathcal{C}_1$ are weakly compact in $L^1(\mu)$. Then $\lim_{\lambda\uparrow 1} D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) = D(\mathcal{C}_1\|\mathcal{C}_0)$.

Proof: Monotonicity in the Rényi order gives $D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) \uparrow L \leq D(\mathcal{C}_1\|\mathcal{C}_0)$ as $\lambda \uparrow 1$ for some $L \in [0, +\infty]$. Suppose that $L < D(\mathcal{C}_1\|\mathcal{C}_0)$ and choose c strictly between them. By Lemma 4, the sets $\mathcal{A}_\lambda = \{(Q, P) \in \mathcal{C}_1 \times \mathcal{C}_0 \mid D_\lambda(Q\|P) \leq c\}$. are weakly closed. They are nonempty, weakly compact, and nested as $\lambda \uparrow 1$. Their intersection is therefore nonempty. Choose $(Q^\star, P^\star)$ in the intersection. Then $D_\lambda(Q^\star\|P^\star) \leq c \quad \forall\lambda < 1$. Letting $\lambda \uparrow 1$ gives $D(Q^\star\|P^\star) \leq c$, contradicting $c < D(\mathcal{C}_1\|\mathcal{C}_0)$. Hence, $L = D(\mathcal{C}_1\|\mathcal{C}_0)$. ■

Appendix B
Exact error exponents on finite alphabets

A. *Lemma 6*

Lemma 6: Let $\mathcal{X}$ be a finite alphabet with $d = |\mathcal{X}|$, let P and Q have full support, and let $0 < r < D(Q\|P)$. Then

$$\lim_{n\to\infty} -\frac{1}{n} \log \beta_n^\star(e^{-nr}; P, Q) = \max_{0<s<1} \frac{1-s}{s} \left[D_s(Q\|P) - r\right].$$

Proof: Set $\psi(s) = \log H_s(Q, P)$. For $0 < s < 1$, the likelihood ratio test with threshold $n[\psi(s) + r]/s$ has Type I error at most e^{-nr} and Type II error at most

$$\exp\left\{-n\frac{1-s}{s}\left[D_s(Q\|P) - r\right]\right\}.$$

Optimising over s proves achievability. For the converse, define

$$R_s(x) = \frac{Q(x)^s P(x)^{1-s}}{H_s(Q, P)}.$$

Direct differentiation gives

$$\frac{d}{ds} D(R_s\|P) = s \operatorname{Var}_{R_s}\left(\log\frac{Q}{P}\right)$$

and

$$\frac{d}{ds}\left\{\frac{1-s}{s}\left[D_s(Q\|P) - r\right]\right\} = \frac{r - D(R_s\|P)}{s^2}.$$

Since $D(Q\|P) > r > 0$, there is a unique $s_r \in (0, 1)$ satisfying $D(R_{s_r}\|P) = r$. It is the maximising order and

$$\max_{0<s<1} \frac{1-s}{s}\left[D_s(Q\|P) - r\right] = D(R_{s_r}\|Q).$$

Fix $0 < t < s_r$, put $V = R_t$, and choose n types $V_n \to V$. After symmetrisation, any admissible test is constant, say c_n, on the type class $\mathcal{T}_{V_n}$. Standard type bounds and the Type I constraint give $c_n \leq (n+1)^d e^{-n[r - D(V_n \| P)]} =: \nu_n$, as $\nu_n \longrightarrow 0$. Hence

$$\mathbb{E}_{Q^{\otimes n}}[1 - \varphi_n] \geq (1 - \nu_n)(n+1)^{-d} e^{-nD(V_n \| Q)}.$$

Taking the infimum over admissible tests, then $n \to \infty$ and $t \uparrow s_r$, gives the matching converse exponent. ■

B. Proof of Theorem 4

Proof: For $\rho \geq 0$, set $\lambda_\rho = 1/(1+\rho)$ and

$$G_r(\rho; P, Q) = \begin{cases} \rho \left[D_{\lambda_\rho}(Q \| P) - r \right], & \rho > 0, \\ 0, & \rho = 0. \end{cases}$$

For $0 < \lambda < 1$, define

$$R_{\lambda; P, Q}(x) = \frac{Q(x)^\lambda P(x)^{1-\lambda}}{H_\lambda(Q, P)}.$$

The identity $(1-\lambda) D(V \| P) + \lambda D(V \| Q) = D(V \| R_{\lambda; P, Q}) - \log H_\lambda(Q, P)$ gives

$$G_r(\rho; P, Q) = \min_{V \in \Delta_d} \left\{ D(V \| Q) + \rho \left[D(V \| P) - r \right] \right\}. \tag{20}$$

The representation shows that G_r is concave in ρ, while joint convexity of D_{λ_ρ} gives joint convexity in (P, Q) for $\rho > 0$ [18, Th. 11]. Uniform full support and compactness give

$$K = \max_{\substack{P \in \mathcal{C}_0 \\ Q \in \mathcal{C}_1}} \max \left\{ D(P \| Q), D(Q \| P) \right\} < +\infty.$$

Since $0 \leq D_\lambda(Q \| P) \leq D(Q \| P)$, G_r is jointly continuous on every compact ρ interval. Choosing $V = P$ in (20) gives $G_r(\rho; P, Q) \leq D(P \| Q) - \rho r \leq K - \rho r$. Choose $\overline{\rho} > K/r$. All relevant maxima may then be restricted to $[0, \overline{\rho}]$. Sion's minimax theorem gives

$$\max_{0 \leq \rho \leq \overline{\rho}} \min_{\substack{P \in \mathcal{C}_0 \\ Q \in \mathcal{C}_1}} G_r(\rho; P, Q) = \min_{\substack{P \in \mathcal{C}_0 \\ Q \in \mathcal{C}_1}} \max_{0 \leq \rho \leq \overline{\rho}} G_r(\rho; P, Q). \tag{21}$$

Let $\rho_r^\star$ maximise the left side and let $(P_r^\star, Q_r^\star)$ minimise the right side. If v_r denotes the common value, then

$$v_r = \min_{P, Q} G_r(\rho_r^\star; P, Q) \leq G_r(\rho_r^\star; P_r^\star, Q_r^\star) \leq \max_\rho G_r(\rho; P_r^\star, Q_r^\star) = v_r.$$

Hence equality holds throughout and the resulting triple is a saddle point. Because $r < D(\mathcal{C}_1 \| \mathcal{C}_0)$, Corollary 6 gives some $\lambda < 1$ such that $D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) > r$. Thus $v_r > 0$. Since $G_r(0; P, Q) = 0$ and $G_r(\overline{\rho}; P, Q) < 0$ for every pair, $0 < \rho_r^\star < \overline{\rho}$. The bound $G_r(\rho; P, Q) \leq K - \rho r$ extends the left saddle inequality to all $\rho \geq 0$.

Set $\lambda_r^\star = 1/(1 + \rho_r^\star)$. Under $\rho = (1-\lambda)/\lambda$, the saddle inequalities become

$$\frac{1-\lambda}{\lambda} \left[D_\lambda(Q_r^\star \| P_r^\star) - r \right] \leq \frac{1-\lambda_r^\star}{\lambda_r^\star} \left[D_{\lambda_r^\star}(Q_r^\star \| P_r^\star) - r \right] \leq \frac{1-\lambda_r^\star}{\lambda_r^\star} \left[D_{\lambda_r^\star}(Q \| P) - r \right] \tag{22}$$

for every $0 < \lambda < 1$, $P \in \mathcal{C}_0$, and $Q \in \mathcal{C}_1$. The right inequality shows that $(Q_r^\star, P_r^\star)$ is projected at order $\lambda_r^\star$. The change of variables in (21) also gives the two variational expressions in (9). The projected bound at $\lambda_r^\star$ gives

$$\liminf_{n \to \infty} -\frac{1}{n} \log \beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \geq G_r(\rho_r^\star; P_r^\star, Q_r^\star).$$

Conversely,

$$\beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1) \geq \beta_n^\star(e^{-nr}; P_r^\star, Q_r^\star).$$

Since $v_r > 0$, $D_{\lambda_r^\star}(Q_r^\star \| P_r^\star) > r$. Monotonicity in the Rényi order therefore gives $D(Q_r^\star \| P_r^\star) > r$. Lemma 6 and the left saddle inequality therefore give

$$\lim_{n\to\infty} -\frac{1}{n} \log \beta_n^\star(e^{-nr}; P_r^\star, Q_r^\star) = G_r(\rho_r^\star; P_r^\star, Q_r^\star).$$

This proves (9) and the asserted attainment. For later use, define

$$R_r^\star(x) = \frac{Q_r^\star(x)^{\lambda_r^\star} P_r^\star(x)^{1-\lambda_r^\star}}{H_{\lambda_r^\star}(Q_r^\star, P_r^\star)}. \tag{23}$$

More generally, let R_ρ denote the same tilted distribution with order λ_ρ. Direct differentiation gives

$$\frac{d}{d\rho} G_r(\rho; P_r^\star, Q_r^\star) = D(R_\rho \| P_r^\star) - r.$$

The derivative vanishes at the interior maximiser, so

$$D(R_r^\star \| P_r^\star) = r. \tag{24}$$

Substitution into (20) gives

$$D(R_r^\star \| Q_r^\star) = \frac{1-\lambda_r^\star}{\lambda_r^\star} \left[D_{\lambda_r^\star}(Q_r^\star \| P_r^\star) - r \right]. \tag{25}$$

■

C. *Proof of Corollary 1*

Proof: For the proof, define $E(r) := \max_{0<\lambda<1} \frac{1-\lambda}{\lambda} \left[D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) - r \right]$ and write $d(\lambda) = D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0)$. Set

$$\rho = \frac{1-\lambda}{\lambda}, \qquad \lambda = \frac{1}{1+\rho},$$

and define

$$\Gamma(\rho) = \rho \, d\left(\frac{1}{1+\rho}\right).$$

Hence

$$E(r) = \sup_{\rho>0} \left\{ \Gamma(\rho) - \rho r \right\}. \tag{26}$$

Fix $0 < r < D(\mathcal{C}_1 \| \mathcal{C}_0)$. The order one limit from below gives $\Gamma(\rho) \to 0$ as $\rho \downarrow 0$, while it also gives a value $\rho_0 > 0$ such that $\Gamma(\rho_0) - \rho_0 r > 0$. These inequalities remain uniform for all rates in a sufficiently small neighbourhood of r. Hence all maximisers are bounded away from zero throughout that neighbourhood.

For the other endpoint, Jensen's inequality gives, for every $P \in \mathcal{C}_0$, $Q \in \mathcal{C}_1$, $(1-\lambda)/(\lambda) D_\lambda(Q \| P) \leq D(P \| Q)$. Where $0 < \lambda < 1$. Compactness and common full support imply

$$K = \sup_{\substack{P\in\mathcal{C}_0 \\ Q\in\mathcal{C}_1}} D(P \| Q) < +\infty.$$

Thus, for rates in a sufficiently small neighbourhood of r, $\Gamma(\rho) - \rho r' \le K - \rho r' \longrightarrow -\infty$ uniformly as $\rho \to \infty$. Therefore all maximisers lie in a common compact interval $0 < \rho_- < \rho < \rho_+ < +\infty$. For a fixed pair (P, Q) define $\gamma_{P,Q}(\rho) = \rho D_{1/(1+\rho)}(Q\|P)$. Since

$$d(\lambda) = \min_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}} D_\lambda(Q\|P),$$

we have

$$\Gamma(\rho) = \min_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}} \gamma_{P,Q}(\rho). \tag{27}$$

Let $\lambda_\rho = 1/(1+\rho)$ and define

$$R_{\lambda_\rho;P,Q}(x) = \frac{Q(x)^{\lambda_\rho} P(x)^{1-\lambda_\rho}}{H_{\lambda_\rho}(Q,P)}.$$

Direct differentiation gives

$$\gamma''_{P,Q}(\rho) = -\lambda_\rho^3 \operatorname{Var}_{R_{\lambda_\rho;P,Q}}\left(\log\frac{Q}{P}\right). \tag{28}$$

Since $D(\mathcal{C}_1\|\mathcal{C}_0) > r > 0$, no distribution in $\mathcal{C}_0$ coincides with one in $\mathcal{C}_1$. The variance in (28) is therefore strictly positive for every pair. Continuity and compactness give $c > 0$ such that $\gamma''_{P,Q}(\rho) \le -c$ uniformly over all pairs and $\rho \in [\rho_-, \rho_+]$. Thus every $\gamma_{P,Q}$ is strongly concave with the same constant. Consequently,

$$\Gamma(\rho) + \frac{c}{2}\rho^2 = \min_{\substack{P\in\mathcal{C}_0\\ Q\in\mathcal{C}_1}} \left[\gamma_{P,Q}(\rho) + \frac{c}{2}\rho^2\right]$$

is concave. Hence Γ is strongly concave and $\rho \longmapsto \Gamma(\rho) - \rho r$ has a unique maximiser $\rho_r^\star$. Therefore

$$\lambda_r^\star = \frac{1}{1+\rho_r^\star}$$

is unique. Danskin's theorem applied to (26) gives

$$E'(r) = -\rho_r^\star = -\frac{1-\lambda_r^\star}{\lambda_r^\star}.$$

Thus the derivative of the exact Type II error exponent is

$$-\frac{1-\lambda_r^\star}{\lambda_r^\star}.$$

Finally, E is convex as the supremum of affine functions of r. Since it is differentiable throughout the open achievable regime, its derivative is continuous there. Hence the exact Type II error exponent is continuously differentiable for $0 < r < D(\mathcal{C}_1\|\mathcal{C}_0)$. ■

D. Proof of Theorem 5

Proof: Write $\Gamma_n(r) := 1 - \beta_n^\star(e^{-nr}; \mathcal{C}_0, \mathcal{C}_1)$. Then

$$\Gamma_n(r) = \sup_{\substack{0\le\varphi_n\le 1\\ \sup_{P\in\mathcal{C}_0}\mathbb{E}_{P^{\otimes n}}\varphi_n \le e^{-nr}}} \inf_{Q\in\mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}\varphi_n. \tag{29}$$

Let $\mathcal{P}_n(\mathcal{X})$ denote the set of n types on $\mathcal{X}$. For $V \in \Delta_d$, define $d_{\mathcal{C}_0}(V) := \inf_{P\in\mathcal{C}_0} D(V\|P)$ and $a_r(V) := [r - d_{\mathcal{C}_0}(V)]_+$. For $Q \in \mathcal{C}_1$, put $F_n(Q) := \min_{V\in\mathcal{P}_n(\mathcal{X})}\{D(V\|Q) + a_r(V)\}$ and $E_n(r) := \sup_{Q\in\mathcal{C}_1} F_n(Q)$. Standard type bounds give, for every n type V and every full support law R,

$$(n+1)^{-d} e^{-nD(V\|R)} \le R^{\otimes n}(T_V) \le e^{-nD(V\|R)}, \tag{30}$$

while $|\mathcal{P}_n(\mathcal{X})| \le (n+1)^d$. For a lower bound on $\Gamma_n(r)$, define

$$\varphi_n(x^n) = (n+1)^{-d} \exp\{-n a_r(\widehat{P}_{x^n})\}. \tag{31}$$

For every $P \in \mathcal{C}_0$,

$$\mathbb{E}_{P^{\otimes n}} \varphi_n \le (n+1)^{-d} \sum_{V \in \mathcal{P}_n(\mathcal{X})} \exp\{-n[D(V\|P) + a_r(V)]\}.$$

Since $D(V\|P) + a_r(V) \ge r$ for every V and $P \in \mathcal{C}_0$, we have $\sup_{P \in \mathcal{C}_0} \mathbb{E}_{P^{\otimes n}} \varphi_n \le e^{-nr}$, so the test is admissible. Fix $Q \in \mathcal{C}_1$ and choose an n type V_Q attaining $F_n(Q)$. The lower type bound gives $\mathbb{E}_{Q^{\otimes n}} \varphi_n \ge (n+1)^{-2d} e^{-nF_n(Q)}$. Taking the infimum over Q gives

$$\Gamma_n(r) \ge (n+1)^{-2d} e^{-nE_n(r)}. \tag{32}$$

For the upper bound, symmetrise any admissible test over coordinate permutations. Its value on a type class T_V may then be written as $t_V \in [0,1]$. Compactness of $\mathcal{C}_0$ gives $P_V \in \mathcal{C}_0$ satisfying $D(V\|P_V) = d_{\mathcal{C}_0}(V)$. The Type I constraint and the lower type bound give $e^{-nr} \ge t_V (n+1)^{-d} e^{-n d_{\mathcal{C}_0}(V)}$. Combining this with $t_V \le 1$ yields

$$t_V \le (n+1)^d e^{-n a_r(V)}. \tag{33}$$

Therefore, for every $Q \in \mathcal{C}_1$,

$$\mathbb{E}_{Q^{\otimes n}} \varphi_n \le (n+1)^d \sum_{V \in \mathcal{P}_n(\mathcal{X})} \exp\{-n[D(V\|Q) + a_r(V)]\} \le (n+1)^{2d} e^{-nF_n(Q)}.$$

Since F_n is continuous on the compact class $\mathcal{C}_1$, its supremum is attained. Choosing a maximising Q and then taking the supremum over admissible tests gives

$$\Gamma_n(r) \le (n+1)^{2d} e^{-nE_n(r)}. \tag{34}$$

Hence

$$(n+1)^{-2d} e^{-nE_n(r)} \le \Gamma_n(r) \le (n+1)^{2d} e^{-nE_n(r)}. \tag{35}$$

Define $F(Q) := \min_{V \in \Delta_d} \{D(V\|Q) + a_r(V)\}$ and $E_{\text{type}}(r) := \sup_{Q \in \mathcal{C}_1} F(Q)$. Uniform full support implies joint continuity of all relevant divergences on the compact domains. Thus $d_{\mathcal{C}_0}(V)$ and $a_r(V)$ are continuous. Since the n types become dense in Δ_d, uniform continuity gives $\sup_{Q \in \mathcal{C}_1} |F_n(Q) - F(Q)| \to 0$, and hence $E_n(r) \to E_{\text{type}}(r)$. Using (35),

$$\lim_{n \to \infty} -\frac{1}{n} \log \Gamma_n(r) = E_{\text{type}}(r). \tag{36}$$

It remains to identify $E_{\text{type}}(r)$. For fixed $Q \in \mathcal{C}_1$, $[x]_+ = \sup_{0 \le a \le 1} ax$ and $r - d_{\mathcal{C}_0}(V) = \sup_{P \in \mathcal{C}_0} [r - D(V\|P)]$. Hence

$$F(Q) = \min_{V \in \Delta_d} \sup_{\substack{0 \le a \le 1 \\ P \in \mathcal{C}_0}} \{D(V\|Q) + a[r - D(V\|P)]\}. \tag{37}$$

Fix $\delta \in (0,1)$ and let $\mathcal{K}_\delta := \{(a,S) \mid 0 \le a \le 1-\delta,\ S = aP,\ P \in \mathcal{C}_0\}$. Convexity and compactness of $\mathcal{C}_0$ make $\mathcal{K}_\delta$ convex and compact. For $a > 0$, put $\Phi_Q(V,a,S) := D(V\|Q) + ar - aD(V\|S/a)$, with its continuous extension at $a=0$. For fixed (a,S), Φ_Q is convex in V. For fixed V, it is concave in (a,S) because $(a,S)\mapsto aD(V\|S/a)$ is the perspective of the convex function $P\mapsto D(V\|P)$. Sion's minimax theorem gives

$$\min_{V}\max_{(a,S)\in\mathcal{K}_\delta}\Phi_Q(V,a,S)=\max_{(a,S)\in\mathcal{K}_\delta}\min_{V}\Phi_Q(V,a,S). \tag{38}$$

For $a\in(0,1)$, set $\lambda:=1/(1-a)>1$, so that $a=(\lambda-1)/\lambda$. A direct variational calculation gives

$$\min_{V\in\Delta_d}\{D(V\|Q)+a[r-D(V\|P)]\}=\frac{\lambda-1}{\lambda}[r-D_\lambda(Q\|P)]. \tag{39}$$

The minimising distribution is proportional to $Q(x)^\lambda P(x)^{1-\lambda}$. Letting $\delta\downarrow 0$ and retaining the $a=0$ branch gives

$$F(Q)=\sup_{P\in\mathcal{C}_0}\sup_{\lambda>1}\frac{\lambda-1}{\lambda}[r-D_\lambda(Q\|P)]_+. \tag{40}$$

Therefore

$$E_{\mathrm{type}}(r)=\sup_{\substack{Q\in\mathcal{C}_1\\P\in\mathcal{C}_0}}\sup_{\lambda>1}\frac{\lambda-1}{\lambda}[r-D_\lambda(Q\|P)]_+\sup_{\lambda>1}\frac{\lambda-1}{\lambda}[r-D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)]_+. \tag{41}$$

Combining (36) and (41) proves (10). The equivalent asymptotic form (12) follows immediately. ■

E. Proof of Lemma 3

Proof: Let $M=\log\Big(\min_{\substack{R\in\mathcal{C}_0\cup\mathcal{C}_1\\x\in\mathcal{X}}}R(x)\Big)^{-1}$. For $P\in\mathcal{C}_0$ and $Q\in\mathcal{C}_1$, let $h_{P,Q}(x)=\log Q(x)/P(x)$. Uniform full support gives $|h_{P,Q}|\le M$, while $\mathbb{E}_P[h_{P,Q}]=-D(P\|Q)$ and $H_\lambda(Q,P)=\mathbb{E}_P\left[e^{\lambda h_{P,Q}}\right]$. Jensen's inequality and Hoeffding's lemma give

$$0\le\log\mathbb{E}_P\left[e^{\lambda h_{P,Q}}\right]-\lambda\mathbb{E}_P[h_{P,Q}]\le\frac{M^2}{2}\lambda^2.$$

Since

$$\frac{1-\lambda}{\lambda}D_\lambda(Q\|P)=-\frac{1}{\lambda}\log H_\lambda(Q,P),$$

we obtain

$$D(P\|Q)-\frac{M^2}{2}\lambda\le\frac{1-\lambda}{\lambda}D_\lambda(Q\|P)\le D(P\|Q).$$

Taking the infimum over $(P,Q)\in\mathcal{C}_0\times\mathcal{C}_1$ and letting $\lambda\downarrow 0$ proves (13). ■

F. Proof of Corollary 2

Proof: For every $P\in\mathcal{C}_0$ and $Q\in\mathcal{C}_1$, pairwise reduction and the Chernoff Stein lemma give

$$\limsup_{n\to\infty}-\frac{1}{n}\log\beta_n^\star(\varepsilon;\mathcal{C}_0,\mathcal{C}_1)\le D(P\|Q).$$

Taking the infimum over the pair gives the converse exponent $D(\mathcal{C}_0\|\mathcal{C}_1)$. For achievability, set$\lambda_n=n^{-1/2}$ and let $M=\log\Big(\min_{\substack{R\in\mathcal{C}_0\cup\mathcal{C}_1\\x\in\mathcal{X}}}R(x)\Big)^{-1}$. Theorem 2 and Lemma 3 give, for $n\ge 2$,

$$-\frac{1}{n}\log\beta_n^\star(\varepsilon;\mathcal{C}_0,\mathcal{C}_1)\ge\frac{1-\lambda_n}{\lambda_n}D_{\lambda_n}(\mathcal{C}_1\|\mathcal{C}_0)-\frac{1-\lambda_n}{n\lambda_n}\log\frac{1}{\varepsilon}\ge D(\mathcal{C}_0\|\mathcal{C}_1)-\frac{M^2}{2\sqrt{n}}-\frac{1-\lambda_n}{n\lambda_n}\log\frac{1}{\varepsilon}.$$

The final two terms vanish, proving (14). If $D(\mathcal{C}_0\|\mathcal{C}_1)=0$, continuity on the compact product and full support give $P_0=Q_0=R$ for some pair attaining the infimum. Every admissible test has Type II error at least $1-\varepsilon$ under $R^{\otimes n}$, while the constant test $\varphi_n\equiv\varepsilon$ attains equality. ■

G. Proof of Corollary 3

Proof: Let $M = \log\left(\min_{\substack{R\in\mathcal{C}_0\cup\mathcal{C}_1\\ x\in\mathcal{X}}} R(x).\right)^{-1}$. Lemma 3 gives, for every $0 < \lambda < 1$,

$$\frac{1-\lambda}{\lambda} D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) \le D(\mathcal{C}_0\|\mathcal{C}_1),$$

which gives the required upper bound. For sufficiently small $r > 0$, take $\lambda = \sqrt{r}$. The lower estimate in the same lemma gives

$$\max_{0<\lambda<1} \frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1\|\mathcal{C}_0) - r\right] \ge D(\mathcal{C}_0\|\mathcal{C}_1) - \frac{M^2}{2}\sqrt{r} - \frac{1-\sqrt{r}}{\sqrt{r}} r = D(\mathcal{C}_0\|\mathcal{C}_1) - \left(\frac{M^2}{2}+1\right)\sqrt{r} + r.$$

Letting $r \downarrow 0$ proves the result. ■

Appendix C
Refining the analysis

A. Proof of Proposition 1

Proof: For the fixed λ and projected pair in proposition 1, define $S^\star(x^n) := \sum_{i=1}^n h_\lambda^\star(x_i)$. Since $\mathcal{X}$ is finite, $S^\star$ takes finitely many distinct values $v_1 < \cdots < v_m$. For $1 \le j \le m$ and $0 \le \eta \le 1$, set

$$\psi_{j,\eta} := \mathbb{1}\{S^\star > v_j\} + \eta\mathbb{1}\{S^\star = v_j\}.$$

Starting from $\psi_{m,0} = 0$, increase η at each boundary in the order $v_m, \dots, v_1$, ending at $\psi_{1,1} = 1$. Adjacent segments join because $\psi_{j,1} = \psi_{j-1,0}$. For each boundary, define

$$A_j(\eta) := \sup_{P\in\mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}\left[\psi_{j,\eta}\right].$$

If $0 \le \eta \le \eta' \le 1$, then $0 \le A_j(\eta') - A_j(\eta) \le \eta' - \eta$. Thus the Type I error is continuous and nondecreasing along the joined chain, with endpoint values zero and one. It therefore has a maximal admissible member whose Type I error is exactly ε. Denote any threshold representation by $(\widehat{\tau}, \widehat{\eta})$. Let

$$\kappa_{\mathcal{C}_0} := \min_{\substack{P\in\mathcal{C}_0\\ x\in\mathcal{X}}} P(x) > 0, \qquad \kappa_{\mathcal{C}_1} := \min_{\substack{Q\in\mathcal{C}_1\\ x\in\mathcal{X}}} Q(x) > 0.$$

If two induced test functions in the chain satisfy $\psi < \psi'$, then for some x_0^n and $\delta > 0$, $\psi'(x_0^n) - \psi(x_0^n) = \delta$. Uniform full support gives

$$\mathbb{E}_{P^{\otimes n}}[\psi'] - \mathbb{E}_{P^{\otimes n}}[\psi] \ge \delta\kappa_{\mathcal{C}_0}^n > 0 \forall P \in \mathcal{C}_0, \qquad \mathbb{E}_{Q^{\otimes n}}[1-\psi] - \mathbb{E}_{Q^{\otimes n}}[1-\psi'] \ge \delta\kappa_{\mathcal{C}_1}^n > 0 \forall Q \in \mathcal{C}_1.$$

Hence the Type I error increases strictly and the Type II error decreases strictly along distinct test functions in the chain. The maximal admissible member is therefore the unique restricted minimiser as a test function. Since it is admissible for the unrestricted problem, (16) follows. ■

B. Corollary 7

Corollary 7: Let $\mathcal{C}_0 = \{P\}$ and $\mathcal{C}_1 = \{Q\}$, where P and Q have full support on a finite alphabet. Then, for every $\lambda \in (0,1)$, $h^\star_\lambda = \log \frac{Q}{P}$, independently of λ. Moreover, for every $n \geq 1$ and $\varepsilon \in (0,1)$, the optimised test $\psi^\star_{\widehat{\tau},\widehat{\eta}}$ from proposition 1 satisfies $\mathbb{E}_{Q^{\otimes n}}\left[1 - \psi^\star_{\widehat{\tau},\widehat{\eta}}\right] = \beta^\star_n(\varepsilon; P, Q)$.

Proof: The only projected pair is $(Q^\star_\lambda, P^\star_\lambda) = (Q, P)$, so $h^\star_\lambda = \log Q/P$ and $S^\star = \log Q^{\otimes n}/P^{\otimes n}$. The projected threshold family is therefore the family of randomised likelihood ratio tests. Its maximal member with Type I error ε is the randomised NP test, which attains $\beta^\star_n(\varepsilon; P, Q)$. ■

C. Lemma 7

Lemma 7: Let $\mathcal{X}$ be finite, let $g : \mathcal{X} \to \mathbb{R}$ be nonconstant, and let $\mathfrak{W}$ be a compact class of full support probability distributions. For i.i.d. observations under $W \in \mathfrak{W}$, set

$$G_n := \sum_{i=1}^{n} g(X_i), \qquad m_W := \mathbb{E}_W[g], \qquad \Delta_{n,W} := G_n - n m_W.$$

For every $0 < s_0 < s_1 < +\infty$, there exists $A < +\infty$ such that, for all $W \in \mathfrak{W}$, $s \in [s_0, s_1]$, $t \in \mathbb{R}$, and $n \geq 1$,

$$\mathbb{E}_{W^{\otimes n}}\left[e^{-s(G_n - t)} \mathbb{1}\{G_n \geq t\}\right] \leq \frac{A}{\sqrt{n}}, \qquad \mathbb{E}_{W^{\otimes n}}\left[e^{s(G_n - t)} \mathbb{1}\{G_n < t\}\right] \leq \frac{A}{\sqrt{n}}.$$

Moreover, for every $0 \leq M_0 < +\infty$, there exist $0 < a_{M_0} < A_{M_0} < +\infty$ and $n_{M_0} < +\infty$ such that, for $n \geq n_{M_0}$ and $|u| \leq M_0 \log n$,

$$a_{M_0} \frac{e^{-su}}{\sqrt{n}} \leq \mathbb{E}_{W^{\otimes n}}\left[e^{-s\Delta_{n,W}} \mathbb{1}\{\Delta_{n,W} \geq u\}\right] \leq A_{M_0} \frac{e^{-su}}{\sqrt{n}},$$
$$a_{M_0} \frac{e^{su}}{\sqrt{n}} \leq \mathbb{E}_{W^{\otimes n}}\left[e^{s\Delta_{n,W}} \mathbb{1}\{\Delta_{n,W} < u\}\right] \leq A_{M_0} \frac{e^{su}}{\sqrt{n}},$$

uniformly over $W \in \mathfrak{W}$ and $s \in [s_0, s_1]$. The same conclusions hold, after changing the constants, when strict and weak endpoints are interchanged.

Proof: Compactness, full support, and nonconstancy of g give $0 < v_0 \leq \mathrm{Var}_W(g) \leq v_1 < +\infty$ and a uniform bound on the centred third absolute moments. The Berry-Esseen inequality [21, Th. 44, p. 2329] therefore gives $B < +\infty$ such that

$$\sup_{\substack{W \in \mathfrak{W} \\ z \in \mathbb{R}}} \left| W^{\otimes n}\left(\frac{G_n - n m_W}{\sqrt{n \, \mathrm{Var}_W(g)}} \leq z \right) - \Phi(z) \right| \leq \frac{B}{\sqrt{n}}.$$

Consequently, for every fixed $L > 0$ there is $A_L < +\infty$ such that

$$\sup_{\substack{W \in \mathfrak{W} \\ y \in \mathbb{R}}} W^{\otimes n}\left(y \leq G_n < y + L\right) \leq \frac{A_L}{\sqrt{n}}.$$

The same estimate holds for endpoint atoms because the Gaussian cdf is continuous and the Berry-Esseen bound also applies to strict cdfs. Partitioning $[t, +\infty)$ and $(-\infty, t)$ into intervals of length L, and summing the resulting geometric series using $s \geq s_0$, proves the first two bounds. For the lower bounds, fix M_0 and write $\sigma_W^2 := \mathrm{Var}_W(g)$. Choose $H > 0$ such that $\frac{H}{\sqrt{2\pi v_1}} > 2B + 2$. For $|u| \leq M_0 \log n$, the normalised endpoints $(u + H)/(\sigma_W \sqrt{n})$ and

$(u+2H)/(\sigma_W\sqrt{n})$ converge to zero uniformly. Hence, for all sufficiently large n, the Gaussian probability of the corresponding interval is at least $(2B+1)/\sqrt{n}$. Berry-Esseen then gives

$$W^{\otimes n}\left(u+H<\Delta_{n,W}\leq u+2H\right)\geq \frac{1}{\sqrt{n}}.$$

On this event, $e^{-s\Delta_{n,W}}\geq e^{-su-2s_1H}$, which proves the first lower bound. The interval $u-2H<\Delta_{n,W}\leq u-H$ gives the second. The corresponding upper bounds follow from the first part with $t=nm_W+u$, after factoring out e^{-su} or e^{su}. The atom estimate permits the stated changes of endpoints. ■

D. Proof of Proposition 2

Proof: Set $\lambda:=\lambda_r^\star$, $P^\star:=P_r^\star$, $Q^\star:=Q_r^\star$, and $h_r^\star:=\log(Q^\star/P^\star)$. Define

$$R(x):=\frac{Q^\star(x)^\lambda P^\star(x)^{1-\lambda}}{H_\lambda(Q^\star,P^\star)}.$$

The proof of Theorem 4 gives $D(R\|P^\star)=r$ and $D(R\|Q^\star)=\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]$, and shows that $(Q^\star,P^\star)$ is projected at order λ. Writing $m:=\mathbb{E}_R[h_r^\star]$, we obtain

$$m=r-\frac{1-\lambda}{\lambda}\left[D_\lambda(Q^\star\|P^\star)-r\right], \tag{42}$$

$$\log H_\lambda(Q^\star,P^\star)=\lambda m-r=-(1-\lambda)m-\frac{1-\lambda}{\lambda}\left[D_\lambda(Q^\star\|P^\star)-r\right]. \tag{43}$$

The proof of Theorem 2 therefore gives $\log\mathbb{E}_P[e^{\lambda h_r^\star}]\leq\log H_\lambda(Q^\star,P^\star)$ for every $P\in\mathcal{C}_0$, and $\log\mathbb{E}_Q[e^{-(1-\lambda)h_r^\star}]\leq\log H_\lambda(Q^\star,P^\star)$ for every $Q\in\mathcal{C}_1$. Define the tilted laws

$$\widehat{P}(x):=P(x)\exp\left\{\lambda h_r^\star(x)-\log\mathbb{E}_P[e^{\lambda h_r^\star}]\right\},$$
$$\widehat{Q}(x):=Q(x)\exp\left\{-(1-\lambda)h_r^\star(x)-\log\mathbb{E}_Q[e^{-(1-\lambda)h_r^\star}]\right\}.$$

The two tilted classes are compact and have uniform full support. Moreover, $h_r^\star$ is nonconstant, since otherwise $P^\star=Q^\star$, contradicting $r<D(\mathcal{C}_1\|\mathcal{C}_0)$. Thus Lemma 7 applies uniformly. With $S_n:=\sum_{i=1}^n h_r^\star(X_i)$, change of measure and (43) give, for every $u\in\mathbb{R}$,

$$P^{\otimes n}(S_n\geq nm+u)\leq C_{\mathcal{C}_0}n^{-1/2}e^{-nr-\lambda u}\qquad\forall P\in\mathcal{C}_0, \tag{44}$$

and

$$Q^{\otimes n}(S_n<nm+u)\leq C_{\mathcal{C}_1}n^{-1/2}\exp\left\{-n\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]+(1-\lambda)u\right\}, \tag{45}$$

uniformly over $Q\in\mathcal{C}_1$. Take $u_n:=-(2\lambda)^{-1}\log n+B$. By (44), the test $\varphi_n^{(B)}:=\mathbb{1}\{S_n\geq nm-(2\lambda)^{-1}\log n+B\}$ has Type I error at most $C_{\mathcal{C}_0}e^{-\lambda B}e^{-nr}$. Choose B so that $C_{\mathcal{C}_0}e^{-\lambda B}\leq 1$. By (42), its threshold is

$$n\left[r-\frac{1-\lambda}{\lambda}(D_\lambda(Q^\star\|P^\star)-r)\right]-\frac{\log n}{2\lambda}+B.$$

Equation (45) then gives

$$\sup_{Q\in\mathcal{C}_1}\mathbb{E}_{Q^{\otimes n}}[1-\varphi_n^{(B)}]\leq C_{\mathcal{C}_1}e^{(1-\lambda)B}n^{-1/(2\lambda)}\exp\left\{-n\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]\right\}.$$

This proves the required upper order. For the converse, put $T_n := S_n - nm$. The exact likelihood ratio identities are

$$(P^\star)^{\otimes n}(A) = e^{-nr}\mathbb{E}_{R^{\otimes n}}\left[e^{-\lambda T_n}\mathbb{1}_A\right],$$
$$(Q^\star)^{\otimes n}(A) = \exp\left\{-n\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]\right\}\mathbb{E}_{R^{\otimes n}}\left[e^{(1-\lambda)T_n}\mathbb{1}_A\right].$$

Applied to $\{R\}$, Lemma 7 gives, for everyfixed $M<+\infty$, uniformly over $|u|\le M\log n$,

$$(P^\star)^{\otimes n}(T_n\ge u) = \Theta\left(e^{-nr}n^{-1/2}e^{-\lambda u}\right),$$
$$(Q^\star)^{\otimes n}(T_n< u) = \Theta\left(n^{-1/2}\exp\left\{-n\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]+(1-\lambda)u\right\}\right),$$

with the same orders for strict or weak endpoints. Consequently, there exist constants $B_-<B_+$ such that, for allsufficiently large n,

$$(P^\star)^{\otimes n}\left(T_n\ge -\frac{\log n}{2\lambda}+B_-\right) > e^{-nr},$$
$$(P^\star)^{\otimes n}\left(T_n> -\frac{\log n}{2\lambda}+B_+\right) < e^{-nr}.$$

Hence the threshold of an exact randomised NP test lies between these two values, and

$$\beta_n^\star(e^{-nr};P^\star,Q^\star) = \Theta\left(n^{-1/(2\lambda)}\exp\left\{-n\frac{1-\lambda}{\lambda}[D_\lambda(Q^\star\|P^\star)-r]\right\}\right).$$

Pairwise reduction gives $\beta_n^\star(e^{-nr};\mathcal{C}_0,\mathcal{C}_1)\ge\beta_n^\star(e^{-nr};P^\star,Q^\star)$, which proves (17). Taking logarithms gives

$$-\frac{1}{n}\log\beta_n^\star(e^{-nr};\mathcal{C}_0,\mathcal{C}_1) = \frac{1-\lambda_r^\star}{\lambda_r^\star}\left[D_{\lambda_r^\star}(Q_r^\star\|P_r^\star)-r\right]+\frac{\log n}{2\lambda_r^\star n}+O\left(\frac{1}{n}\right).$$

■

E. *Proof of Corollary 4*

Proof: By Corollary 1,

$$\frac{d}{dr}\max_{0<\lambda<1}\frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)-r\right] = -\frac{1-\lambda_r^\star}{\lambda_r^\star}.$$

Therefore

$$\frac{1}{2}\left[1-\frac{d}{dr}\max_{0<\lambda<1}\frac{1-\lambda}{\lambda}\left[D_\lambda(\mathcal{C}_1\|\mathcal{C}_0)-r\right]\right] = \frac{1}{2}\left[1+\frac{1-\lambda_r^\star}{\lambda_r^\star}\right] = \frac{1}{2\lambda_r^\star},$$

which proves the result. ■

APPENDIX D
LEAST FAVOURABILITY

A. *Proof of Proposition 3*

Proof: For each $n\ge1$, write $S^\star(x^n):=\sum_{i=1}^n h_\lambda^\star(x_i)$. The two ordering assumptions in Proposition 3, together with preservation of stochastic order under independent sums, give

$$S^\star \text{ under } P^{\otimes n}\le_{\rm st} S^\star \text{ under } (P_\lambda^\star)^{\otimes n},\qquad S^\star \text{ under } Q^{\otimes n}\ge_{\rm st} S^\star \text{ under } (Q_\lambda^\star)^{\otimes n}.$$

The function $s\mapsto\mathbb{1}\{s>\tau\}+\eta\mathbb{1}\{s=\tau\}$ is nondecreasing, while its miss function is nonincreasing. Therefore

$$\mathbb{E}_{P^{\otimes n}}[\psi_{\tau,\eta}^\star]\le\mathbb{E}_{(P_\lambda^\star)^{\otimes n}}[\psi_{\tau,\eta}^\star],\qquad \mathbb{E}_{Q^{\otimes n}}[1-\psi_{\tau,\eta}^\star]\le\mathbb{E}_{(Q_\lambda^\star)^{\otimes n}}[1-\psi_{\tau,\eta}^\star].$$

Since the projected distributions belong to their respective classes, taking the suprema gives the two equalities in proposition 3. ■

B. Proof of Corollary 5

Proof: Every test satisfying the composite Type I constraint is admissible for the selected simple pair. Hence

$$\beta_n^\star(\varepsilon;\mathcal{C}_0,\mathcal{C}_1) \geq \beta_n^\star(\varepsilon;P_\lambda^\star,Q_\lambda^\star).$$

Conversely, the first equality in proposition 3 makes $\psi^\star_{\tau,\eta}$ admissible for the composite problem, while the second identifies its composite Type II error with its Type II error under $Q_\lambda^\star$. Its assumed optimality for the simple pair therefore gives the reverse inequality. ■

C. Lemma 8

Lemma 8: Let $\{P_\theta : \theta \in \Theta\}$ be the natural exponential family introduced in Section VI. For every $n \geq 1$, $\theta_1 < \theta_2$, and bounded measurable nondecreasing $f : \mathbb{R} \to \mathbb{R}$,

$$\mathbb{E}_{P_{\theta_1}^{\otimes n}}[f(T_n)] \leq \mathbb{E}_{P_{\theta_2}^{\otimes n}}[f(T_n)], \qquad T_n := \sum_{i=1}^n T(X_i).$$

Proof: For $\theta_2 > \theta_1$,

$$\frac{dP_{\theta_2}^{\otimes n}}{dP_{\theta_1}^{\otimes n}} = \exp\{(\theta_2-\theta_1)T_n - n[\psi(\theta_2)-\psi(\theta_1)]\} =: L(T_n),$$

where L is nondecreasing and has expectation one under $P_{\theta_1}^{\otimes n}$. If U, U' are independent copies of T_n under this law, then $2\,\mathrm{Cov}(f(U),L(U)) = \mathbb{E}[(f(U)-f(U'))(L(U)-L(U'))] \geq 0$. Since the difference of the two expectations in the lemma equals this covariance, the claim follows. ■

D. Proof of Proposition 4

Proof: Suppose first that $\theta_+^{\mathcal{C}_0} < \theta_-^{\mathcal{C}_1}$. The likelihood ratio $dP_{\theta_-^{\mathcal{C}_1}}/dP_{\theta_+^{\mathcal{C}_0}}$ is increasing in T. By the NP lemma, an optimal endpoint test has the form $\varphi_n^{\mathrm{e}} = \mathbb{1}\{T_n > t\} + \eta\mathbb{1}\{T_n = t\}$, for $0 \leq \eta \leq 1$. Lemma 8 and monotonicity of this test and its miss function give

$$\sup_{P\in\mathcal{C}_0} \mathbb{E}_{P^{\otimes n}}[\varphi_n^{\mathrm{e}}] = \mathbb{E}_{(P_{\theta_+^{\mathcal{C}_0}})^{\otimes n}}[\varphi_n^{\mathrm{e}}] \leq \varepsilon, \qquad \text{and} \qquad \sup_{Q\in\mathcal{C}_1} \mathbb{E}_{Q^{\otimes n}}[1-\varphi_n^{\mathrm{e}}] = \mathbb{E}_{(P_{\theta_-^{\mathcal{C}_1}})^{\otimes n}}[1-\varphi_n^{\mathrm{e}}].$$

Thus the endpoint test is admissible for the composite problem and attains the simple endpoint error. Every test admissible for the composite problem is also admissible for the endpoint pair, giving the reverse inequality. If $\theta_+^{\mathcal{C}_1} < \theta_-^{\mathcal{C}_0}$, the same argument applied to $-T$ gives the lower-tail endpoint reduction. It remains to identify the projected pair. For $0 < \lambda < 1$,

$$D_\lambda\left(P_{\theta_{\mathcal{C}_1}} \middle\| P_{\theta_{\mathcal{C}_0}}\right) = \frac{1}{\lambda-1}\Big[\psi(\lambda\theta_{\mathcal{C}_1} + (1-\lambda)\theta_{\mathcal{C}_0}) - \lambda\psi(\theta_{\mathcal{C}_1}) - (1-\lambda)\psi(\theta_{\mathcal{C}_0})\Big].$$

Differentiation gives

$$\frac{\partial D_\lambda}{\partial\theta_{\mathcal{C}_0}} = \psi'(\theta_{\mathcal{C}_0}) - \psi'(\lambda\theta_{\mathcal{C}_1} + (1-\lambda)\theta_{\mathcal{C}_0}), \qquad \frac{\partial D_\lambda}{\partial\theta_{\mathcal{C}_1}} = \frac{\lambda}{\lambda-1}\left[\psi'(\lambda\theta_{\mathcal{C}_1} + (1-\lambda)\theta_{\mathcal{C}_0}) - \psi'(\theta_{\mathcal{C}_1})\right].$$

Since ψ' is strictly increasing, when $\theta_{\mathcal{C}_0} < \theta_{\mathcal{C}_1}$ the divergence decreases with $\theta_{\mathcal{C}_0}$ and increases with $\theta_{\mathcal{C}_1}$. Thus the minimum occurs at $\theta_{\mathcal{C}_0} = \theta_+^{\mathcal{C}_0}$ and $\theta_{\mathcal{C}_1} = \theta_-^{\mathcal{C}_1}$, so the projected pair is $(P_{\theta_-^{\mathcal{C}_1}}, P_{\theta_+^{\mathcal{C}_0}})$. When the order of the intervals is reversed, the derivative signs reverse and the projected pair is $(P_{\theta_+^{\mathcal{C}_1}}, P_{\theta_-^{\mathcal{C}_0}})$. These are exactly the endpoint pairs stated in the proposition. ∎

APPENDIX E
NUMERICAL EVALUATION OF THE BOUNDS

This appendix gives details of the numerical optimisation used in Section VII.

For the affine classes $\mathcal{C}_0 = \{P_s : s \in [0,1]\}$, and $\mathcal{C}_1 = \{Q_t : t \in [0,1]\}$, the composite Rényi divergence is evaluated by the joint minimisation

$$D_\lambda(\mathcal{C}_1 \| \mathcal{C}_0) = \min_{(s,t)\in[0,1]^2} D_\lambda(Q_t \| P_s). \tag{46}$$

Thus, for each fixed λ, the two class parameters are optimised jointly over $[0,1]^2$. We first evaluate the objective over the parameter square to locate candidate minima and then refine the best candidates by constrained numerical optimisation. The calculation is repeated from multiple starting points and checked on a finer parameter grid. The same procedure is used for $D_\lambda(\mathcal{C}_0 \| \mathcal{C}_1)$ when the opposite divergence direction is required.

For the exponentially decreasing Type I constraint $\varepsilon_n = e^{-nr}$, Theorem 3 gives the critical rate

$$r_{\mathrm{c}} = D(\mathcal{C}_1 \| \mathcal{C}_0) = \min_{(s,t)\in[0,1]^2} D(Q_t \| P_s). \tag{47}$$

For the first affine ternary example, $r_{\mathrm{c}} = 0.094$, $s^\star = 0$, and $t^\star = 0.639$. The rates used in Fig. 1 are $r = 0.35 r_{\mathrm{c}} = 0.033$ and $r = 1.5 r_{\mathrm{c}} = 0.142$.

For the final affine ternary example, $D(\widetilde{\mathcal{C}}_1 \| \widetilde{\mathcal{C}}_0) = 0.017$, and $D(\widetilde{\mathcal{C}}_0 \| \widetilde{\mathcal{C}}_1) = 0.016$. For the achievable regime, we optimise the expression in Equation (9). The optimisation over the Rényi order is performed outside the joint class optimisation. Thus, for every trial value $0 < \lambda < 1$, we first compute

$$(s_\lambda^\star, t_\lambda^\star) \in \operatorname*{arg\,min}_{(s,t)\in[0,1]^2} D_\lambda(Q_t \| P_s),$$

and then optimise the resulting scalar function of λ.

For the first affine ternary example with $r = 0.35 r_{\mathrm{c}}$, this gives $\lambda^\star = 0.601$, $s_{\lambda^\star}^\star = 0$, and $t_{\lambda^\star}^\star = 0.602$, with $D_{\lambda^\star}(\mathcal{C}_1 \| \mathcal{C}_0) = 0.056$.

For the converse regime, we optimise the expression in Equation (10). Again, for every trial value $\lambda > 1$, the class parameters are first optimised jointly over $[0,1]^2$. For the first affine ternary example with $r = 1.5 r_{\mathrm{c}}$, this gives $\lambda^\star = 1.224$, $s_{\lambda^\star}^\star = 0$, and $t_{\lambda^\star}^\star = 0.662$, with $D_{\lambda^\star}(\mathcal{C}_1 \| \mathcal{C}_0) = 0.116$.

For a general Type I constraint ε_n, the reverse finite sample converse is the bound in Theorem 1. For the numerical figures we also use the converse obtained from the opposite divergence direction,

$$B_{n,\mathrm{fwd}} = \exp\left\{ \sup_{\lambda>1} \left[\frac{\lambda}{\lambda-1} \log(1-\varepsilon_n) - n D_\lambda(\mathcal{C}_0 \| \mathcal{C}_1) \right] \right\}. \tag{48}$$

The displayed Rényi converse is the larger of this bound and the bound in Theorem 1. For each Rényi order, the corresponding composite divergence is obtained by joint optimisation over $(s,t) \in [0,1]^2$ before the outer optimisation over λ.

For the fixed and subexponential Type I constraints in Fig. 4, we use the calibrated projected test of Proposition 1. Since the projected pair depends on λ, for each candidate order $0 < \lambda < 1$ we first compute

$$(s_\lambda^\star, t_\lambda^\star) \in \arg\min_{(s,t)\in[0,1]^2} D_\lambda(\widetilde{Q}_t \| \widetilde{P}_s).$$

The corresponding projected log likelihood ratio is then formed and its threshold and boundary randomisation are calibrated under the required Type I constraint. The resulting Type II error is evaluated over the complete alternative class, and the order giving the smallest value is selected. We use 27 candidate Rényi orders between 0.001 and 0.99, with increased resolution near zero and one. The converse orders are instead obtained by continuous one dimensional optimisation over $\lambda > 1$.

All numerical calculations were implemented in Python using NumPy, SciPy, and Matplotlib. NumPy was used for array operations and polynomial representations, while SciPy was used for numerical optimisation, root finding, special functions, and the linear programmes through its HiGHS interface. The optimisation routines included constrained local optimisation, scalar minimisation, Brent root finding, and differential evolution for independent global checks. Matplotlib was used to generate the numerical figures, and independent blocklength calculations were parallelised using Python's `concurrent.futures` module.

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